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State-Building at What Cost? The Consequences of Military Control in Medieval China

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Abstract

A key challenge to state-building in conflict-prone societies is the proliferation of autonomous armed actors. As governments seek to enhance their monopoly over violence, consequences of military centralization deserve scholarly attention. We argue that centralization undermines state's coercive capacity: by weakening the leaders of armed groups, it simultaneously erodes the military efficiency and discipline of these groups, which also serve as agents of coercion for the state. We leverage a centralization reform in China's Tang Dynasty (618-907 CE) after a civil war, using unique data on soldier mutinies, rebellions by local generals, civilian uprisings, and battle outcomes. Difference-in-differences estimates reveal that centralization successfully reduced local generals' rebellions, but led to increased soldier mutinies against the generals and more civilian uprisings. It also worsened battlefield performance. While research on state capacity has focused on its extractive and administrative dimensions, these findings deepen our understanding about another important aspect: the coercive dimension.

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A weak center presiding over strong local powerholders has been a prevalent political setting in human history and the developing world today (Finkel & Gehlbach, 2020; Garfias, 2018; Garfias & Sellars, 2021; Migdal, 1988). In conflict-prone societies, local powerholders often manifest themselves as autonomous armed actors that proliferate across territories, posing acute challenges to political order (e.g., Esberg, 2024; Lessing, 2017; Wang, 2022, pp. 177–200).

While coercive power forms a key aspect of state capacity (Hanson & Sigman, 2021), current work on state-building pays limited attention to how states seek to consolidate control over coercive agents, and how such efforts reshape the state's capacity to maintain order and defend against threats. Focusing on coercive agents also sheds new light on why state-building can backfire: centralization may provoke threatened elites (Chiovelli et al., 2023; Garfias & Sellars, 2022), but it can also unsettle the local chains of command through which coercive actors maintained order.

This chain-of-command problem resonates with the guardianship dilemma in civil–military relations (e.g., Feaver, 1999; McMahon & Slantchev, 2015; Paine, 2022; Powell, 2014), but places it within the bureaucratic organization of local armed forces, where military control works through personnel authority: by reallocating appointment and promotion power, the center shifts subordinates' career dependence away from local commanders and toward itself. This administrative intervention treats local militaries as internally differentiated organizations, where loyalty, coordination, and authority can be reshaped from within rather than only through the center's direct confrontation with local powerholders.

This internal reorganization underpins the core trade-off we study. While military centralization strengthens the center's control over local military powerholders and reduces their acts of violence, it can also undermine coercive capacity by weakening commanders' authority over the troops who fight and police on behalf of the state. The same personnel controls that constrain local powerholders can therefore compromise military effectiveness, discipline, and internal cohesion, creating openings for mutiny from below. To empirically explore this argument, we leverage a military centralization reform in the latter half of China's Tang Dynasty (618–907 CE), a historical episode in which a weak central government ruled over rebellious military powerholders in the localities.

Between 755 and 763 CE, Tang China survived a highly destructive civil war. As a result, newly-armed, semi-autonomous "commissioners" proliferated across the Chinese hinterland, serving simultaneously as the civil and military governors of Tang provinces. Often contesting with the center for greater local autonomy, they were a constant source of instability. In 819 CE, the central government responded with a military centralization reform (hereafter "the Reform") to enhance central control over the commissioners' armed forces. Under this reform, commissioners only retained full command of armies in the provincial headquarters. Command over armies in the non-headquarter, "outer" prefectures[1] was given to centrally-appointed prefects: in other words, authority over parts of the forces within the powerholders' jurisdictions became centralized at the hands of the emperor.[2]

We hand-collect records on commissioner-controlled armies (abbreviated as "CCA") in the outer prefectures in the pre-Reform period from a variety of sources including official histories, imperially-commissioned works on politics and history, political writings by Tang officials, and tomb epitaphs of Tang elites. We also geocode datasets on outcomes of political violence, including over 200 instances of commissioner-led rebellions, soldier-led mutinies and civilian uprisings, and 420 battles fought by commissioners on behalf of the Tang government. Lastly, we compile biographical information for 1,450 military commissioners. As the Reform transferred CCAs in outer prefectures from commissioner control to the hands of central government agents, we exploit variation in the pre-Reform number of CCA prefectures in a province as a measure of reform exposure. A difference-in-differences strategy estimates the causal impact of military centralization on outcomes of military control and coercive capacity.

We document that provinces with higher exposure to military centralization had a reduced likelihood of commissioner-led rebellions against the center in the post-Reform period. This finding reflects enhanced military control over the local powerholders, as commissioners headed the highest level of subnational administration in Tang China. As another indicator of control, the Reform also substantially increased the frequency with which the commissioners departed their provinces due to centrally-mandated reappointment.

We then demonstrate that, after the Reform, those provinces also experienced an increased likelihood of soldier-led mutinies against the commissioner, which occurred in the provincial headquarters, and an increased likelihood of civilian uprisings. These results highlight a decline in the state's capacity to enforce political order. Last, we present evidence that those provinces had worse performance in battles against the regime's enemies, signaling a decrease in the state's coercive ability.

Using quantitative and qualitative evidence, we illustrate that our findings on civilian uprising and battle performance were likely driven by communication and coordination problems that became exacerbated by military centralization. Moreover, increased soldier mutiny within the headquarters army likely resulted from the weakening of commissioner's command over the outer prefecture armies, which functioned as an external constraint on the headquarters army. We highlight military corporate interests (Grewal 2019, 2024) as another factor underlying soldier mutinies. Auxiliary analyses provide evidence against alternative explanations centering on the interplay between mutiny and civilian uprising, quality of governance, commissioner characteristics and re-appointments, participation in centrally-mandated battles, and changes in economic or fiscal conditions.

Theoretical Contributions

Recent research generally decomposes state capacity into three interrelated dimensions: extractive, administrative, and coercive (e.g., Cingolani et al., 2015; Hanson & Sigman, 2021; Wang & Xu, 2018). While most works examine the strategies and consequences of state-building in the first two dimensions,[3] our paper is among the first to study how state-building in the military sphere affects coercive capacity by altering the dynamics of authority among coercive agents.

Importantly, our context of state-building and military centralization against local powerholders contrasts with and extends the civil-military relations literature on coup-proofing and the guardianship dilemma, which primarily examines strategic tensions between the ruler and national armies stationed in the capital region. Our findings thus also speak to modern contexts of violence management where governments outsource security tasks to semi-autonomous violent actors (Carey & Mitchell, 2017; Staniland, 2021), and our results on rebellion and battle outcomes enrich existing evidence on how coup-proofing affects coup occurrence and military effectiveness (e.g., Belkin & Schofer, 2003; Böhmelt & Pilster, 2015; Narang & Talmadge, 2018; Pilster & Böhmelt, 2011; Powell, 2012; Roessler, 2011).

Our study also contributes to understanding how military control reshapes not only the relationship between ruler and commander, but the internal loyalty and coordination structures of the commander's own forces. Specifically, the reform we study highlights personnel control as a distinct strategy of military control alongside counterbalancing (Brooks, 2019; De Bruin, 2018) and the exploitation of personalist (family, ethnic, or sectarian) and economic ties (e.g., Bellin, 2012; Brooks, 1998; Quinlivan, 1999). Rather than creating rivalrous units or recruiting from specific groups, personnel

control operates as a bureaucratic procedure that institutionally exacerbates the divergence in incentives between higher commanders and their subordinates, redirecting the latter's loyalty toward the ruler. This mechanism relates to recent work that situates civil–military relations within the broader context of centralized bureaucracy (Jost & Mattingly, 2026; Peng, 2027).

Viewing the local military as a non-monolithic organization of actors with varying preferences (Albrecht & Eibl, 2018), we advance a small but growing literature on mutinies that has primarily focused on soldier grievances (Dwyer, 2018; Johnson, 2021). We theorize an alternative mechanism where incentive changes weaken subordinates' willingness to defend their commander's interests, allowing latent commander-subordinate conflict to escalate into intra-military power struggles. Our findings thus illustrate how military control can open up opportunities for mutinies by altering the military's *internal* dynamics (Albrecht, 2015; Nassif, 2015b; Nepstad, 2013).

State-Building and Coercive State Capacity

We present a theoretical framework of state-building in the military sphere and its impact on coercive state capacity, centering on the internal dynamics of local militaries.

Following the Weberian tradition which views the state as a monopoly over violence, we define "state-building through military centralization"[4] as efforts by the central government to consolidate its direct control over the armed forces of local military powerholders. This definition finds parallels in recent literature on historical political economy that treats state-building in the extractive dimension as consolidation of power and revenue under a central authority (Garfias & Sellars, 2024). Across developing societies, local military powerholders are prevalent, and can take various forms ranging from leaders who rose to power through interstate and civil wars (e.g., Berman, 2010; Wang, 2022) to regional armed groups that hold territorial control (e.g., Henn et al., 2025).

State-building through military centralization may involve enhanced surveillance over the powerholder, direct control over branches of the powerholder's armed forces by representatives of the central government, or direct control by the center over military personnel within the powerholders' purview. The scope conditions for credible enforcement of such measures would require the center to be strong enough to deter coordinated resistance by the targeted powerholders. Relatedly, resistance would be less likely to transpire if powerholders' losses are acceptable so that acquiescence is rational.

As a coercive apparatus, military organizations underpin two aspects of state power: the state's monopoly over violence, which refers to its exclusive authority over the use of force (Weber, 1919); and *coercive capacity*, which refers to its ability to effectively deploy forces to protect against external threats and maintain internal order (Hanson & Sigman, 2021; Tilly, 1990). The role of local military powerholders is twofold. They can exercise coercive capacity on behalf of the state by combating internal and external enemies, or challenge the state's monopoly over violence by using violence to pursue centrifugal objectives. The central government thus has incentives to engage in military centralization to constrain those powerholders as alternative perpetrators of violence.

Dynamics in the military shape how the state's coercive capacity changes when the central government enhances control over them. The powerholder and his subordinates, as distinct actors in the military organization, can harbor different preferences and objectives (Albrecht & Eibl, 2018). Officers and soldiers may have their own "corporate interests," derived from their privileged economic and political positions (Finer, 1974; Needler, 1966) or from their tendency to develop a specific collective identity (Janowitz, 1976). Similarly, concerns over career prospects can affect the subordinates' compliance with the powerholder.

Negative Impacts on Coercive Capacity

We argue that state-building through military centralization, while strengthening the state's monopoly over violence, can produce negative impacts on its coercive capacity. We highlight two mechanisms: communication and coordination, and incentive divergence.

By increasing the center's control over the powerholder's armed forces, state-building can disrupt communication and coordination between the powerholder and his subordinates. The central government can impose its own preferences, which often differ from the powerholder's,[5] over the types of individuals selected for subordinate positions. This would empower individuals who are personally and professionally distant from the powerholder, thereby reducing the shared knowledge and tacit understanding and increasing information and communication frictions between the two sides. Such frictions could impede effective coordination and collaboration within the military organization[6] and reduce the army's ability to apply its power—i.e., to organize and deploy its units to execute combat strategies.

Now, we discuss how the above mechanism relates to the state's monopoly over violence and coercive capacity. We regard (1) powerholder-led rebellion against the center as direct challenges to the state's monopoly over violence.

We then focus on three manifestations of coercive capacity, each capturing a different function of the coercive apparatus: (2) battlefield performance, which reflects the military's ability to defend against external threats; (3) civilian uprising, which reflects its ability to deter unrest and maintain internal social order; and (4) soldier-led mutiny against the powerholder, which reflects military discipline—the cohesion of the coercive apparatus underpinning both external defense and internal order.

The decline in military effectiveness, induced by communication and coordination problems, would translate into worse battlefield performance in general. For the powerholder, this would likewise decrease his success likelihood of armed rebellion against the center, thereby weakening his incentives to stage a rebellion. In other words, military centralization can strengthen the state's monopoly over violence by limiting the use of unsanctioned violence.

On the other hand, worsened battlefield performance, itself a decline in coercive capacity, also lowers the military's ability to deter and contain civilian uprisings. Holding constant the level of grievance in the local society, uprisings may increase in number if potential rioters become less deterred by an incompetent military as their expectations of a well-executed repression decrease. They may also increase in scale if the military cannot crush them immediately, thereby allowing them to spread across space. Taken together, military centralization may erode coercive capacity at battles, and also at enforcing political order by loosening the state's control over the grassroots society.

Furthermore, reduced coordination may also provide opportunities (Finer, 1974; Huntington, 1968) for pre-existing conflicts of interest *within* the military organization to escalate into mutinies against the powerholder. As the powerholder's ability to coordinate a defense among his subordinates becomes lower, his opposition in the military would be more capable of staging mutinies to advance their own corporate interests (Brooks, 2013; Grewal, 2019). Thus, state-building may diminish military discipline—i.e., cohesion of the military—by weakening the deterrence on the powerholder's subordinates. Such breakdowns in internal military discipline represent another way in which military centralization may erode the state's coercive capacity.

Beyond communication and coordination, military centralization introduces a second mechanism: incentive divergence between the powerholder and his subordinates.[7] As the center acquires greater authority over the subordinates, it can redirect their career dependence away from the powerholder and towards the central government through personnel control.[8] When their positions and promotion prospects are tied to the state rather than to the powerholder, the subordinates become less committed to the latter's

interests and causes, especially if they conflict with the interests of the former. This weakened commitment would further reduce the subordinates' willingness to participate in powerholder-led rebellions against the center, and to defend the powerholder against internal opposition (i.e., soldier-led mutinies) since doing so no longer carries a clear reward. Whereas the coordination and communication mechanisms weakens the powerholder's ability to mobilize and command his forces, incentive divergence weakens the subordinates' motivation to comply as they have no personal stake in his survival.

Taken together, our theory predicts that state-building, in the form of centralizing control over local powerholders' armed forces, would:

(1) Strengthen the state's monopoly over violence, by reducing powerholders' acts of rebellion.
(2) Lower the state's coercive capacity, reflected in worse battlefield performance (defense against threats), weakened military discipline (more mutinies against the powerholder), and reduced deterrence of civilian uprisings.

Historical Background

This section describes the provincial military commissioners—the "local military powerholders"—in the Tang dynasty, and the military centralization reform of 819 CE.

Tang Warfare and the Origin of Commissioners

The Tang dynasty of China is known for its unprecedented territorial expansion. Its armies usually consisted of infantry and cavalry, and required significant communication and coordination across different units on the battlefield (Graff, 2003, pp. 193, 254).[9] On its vast frontiers, the Tang faced incessant threats from peoples such as the Tibetans, Turks, and Khitans. By 742 CE, professionalized armies were permanently stationed across frontier provinces to sustain defense. These forces were headed by military commissioners, many of whom also wielded considerable civil authority, posing a significant threat to the center (Lewis, 2012, p. 48).

In 755 CE, An Lushan, a military commissioner in the northeast, rebelled and declared himself the emperor. Severe fighting erupted across the Chinese hinterland, and the ensuing eight-year civil war inflicted catastrophic damage on the empire. Though the dynasty survived, a pivotal legacy of this watershed, the focus of this paper, was the proliferation of semi-autonomous military powerholders.

The Post-Conflict Setup: Central Authority Gravely Weakened

Since 755 CE, military commissioners were established across the Chinese hinterland to combat rebels. Granted full discretion over local troops, they proved difficult for emperors to remove post-rebellion. By 763 CE, the northeastern provinces (Hebei) became completely independent.[10] Most other provinces, while nominally under imperial control, were governed by commissioners who exercised both military and administrative authority, developing forms of autonomy that were troubling for the center (Graff, 2003, p. 229; Peterson, 1979, p. 485).

As *local military powerholders*, commissioners commanded large, professional standing armies recruited from the provinces. While tasked with executing imperial military orders,[11] heavy militarization in the provinces also fueled their secessionist or even imperial ambitions, causing decades of political crises. Confrontations between Emperor Dezong and recalcitrant provinces in 781 CE ignited widespread rebellions that lasted until 786 CE. Forced to flee the capital, apologize, and pardon the rebels, Dezong subsequently adopted a "policy of indulgence" towards the commissioners until his death in 805 CE (Peterson, 1979, pp. 506–10).

Internal Structure of Provincial Armies. In the post-755 administrative hierarchy, a provincial commissioner presided over prefectures, each led by an emperor-appointed prefect (Figure 1(a)). The commissioner and his headquarters army were stationed in the headquarters prefecture,[12] while garrison armies were located in non-headquarters, or "outer," prefectures (Graff, 2003, p. 230; Zhang, 2010, p. 84). The commissioner controlled the headquarters and outer prefecture armies, which were commanded respectively by himself and by subordinates that he appointed (Chen, 2005).

Before 819 CE, the stationing of a commissioner-controlled army (CCA) in outer prefectures varied across provinces. This variation, which our empirical strategy exploits, primarily arose from military exigencies during the An Lushan Rebellion. To crush the rebels and other insurgencies, Tang rulers granted local powerholders the discretion to raise troops where needed (Graff, 2003, p. 229; Li, 2015, pp. 380–530; Zhang, 2010, p. 78; Zhang, 2011, p. 101), resulting in garrison armies in certain outer prefectures (see Appendix Section D.2).

The headquarters army was the province's largest force, capable of suppressing a rebellious CCA. However, as the "ultimate guarantee" of a commissioner's power, it also posed the greatest threat of mutiny (Graff, 2003, pp. 230–31). The commissioner controlled their military pay (Zhang, 2010, pp. 64–71).[13] Exploiting their proximity to the commissioner and his

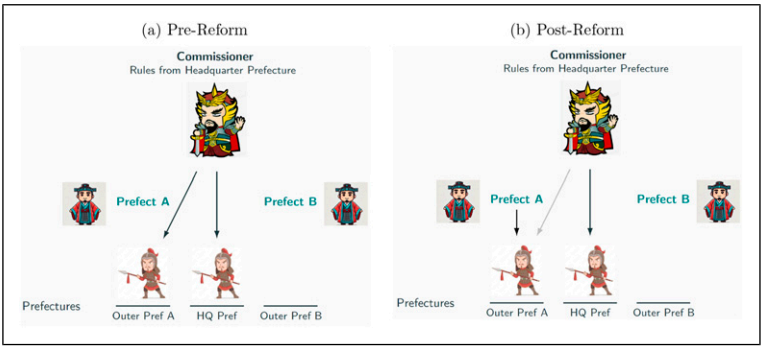


Figure 1. Internal structure of provincial armies, before and after reform.
Note. The figures provide a graphical illustration of the internal structure of provincial armies. Black arrow indicates direct control. Outer prefecture A has a CCA, and is therefore a CCA prefecture in our analysis. Every outer prefecture is headed by a prefect, who is appointed by the emperor. Before the Reform, the commissioner directly controls CCAs in CCA prefectures; after the Reform, the local prefect directly controls CCAs, and the commissioner can indirectly command CCAs through the prefects under central mandate. Armies in the headquarters prefecture ("HQ Pref") are unaffected by the Reform

dependence on their loyalty, they increasingly demanded higher salaries and privileges for maintaining allegiance (Graff, 2003, pp. 231–39). These "unruly soldiers" (*jiaobing*) developed a collective identity, shaped by their shared background as landless peasants, localized urban interests, and semi-hereditary military status (Graff, 2003, pp. 238–239; Li, 2015, pp. 108–109). Their mutinous behaviors and motives are comparable to their modern counterparts in West and Central Africa (Dwyer, 2018), and their anger over general mistreatment often reflected deeper grievances with the political structure.

The Reform of 819 CE

Reasserting central authority over the local military powerholders after the An Lushan Rebellion was a top imperial priority (Graff, 2003, p. 233). To this end, the Reform of 819 CE was deemed "the most significant measure… altering the lines of military authority in the provinces" (Peterson, 1979, p. 535).

The Reform centralized military power by transferring control of outer prefecture armies from commissioners to local prefects (Figure 1(b)).[14] Personnel authority over the prefects was held by the Emperor—they were centrally appointed and managed and exercised military control on the

Emperor's behalf (Chen, 2005, p. 173; Yan, 2023, p. 164).[15] Consequently, after the Reform, only the headquarters army remained under commissioners' direct command (Lewis, 2012, p. 65), while the rest became centralized through agents of the Emperor. The aim was to weaken the commissioners' ability to unilaterally mobilize provincial forces.[16]

In essence, the Reform expanded central control over provincial armies at the expense of their otherwise coherent command structure. Previously, CCA commanders were the commissioner's military men;[17] now they were prefects, who were not members of the local army, had no personal ties to the commissioner, and were accountable to the Emperor. In line with the communication and coordination mechanisms in the Theory Section, it became more difficult for commissioners "to coordinate the many prefectural units" (Dalby, 1979, p. 619).

The Reform also altered incentives. Since prefects' careers depended on the Emperor, they had little reason to join a commissioner's revolt. This also meant outer prefecture armies were less likely to defend a commissioner from a mutiny in the headquarters, making such mutinies more probable.[18] Our theory therefore predicts a decrease in battle performance and commissioner-led rebellions, but an increase in civilian uprisings and mutinies.

Scope Conditions. The Reform met little resistance in 819 CE, and satisfied the two scope conditions discussed in 1. First, at the time of the Reform, Emperor Xianzong had secured several decisive victories against rebellious provinces, which simultaneously strengthened the center's authority and fractured the coalitions that might have organized collective resistance.[19] At the subordinate level, commanders of outer-prefecture CCAs who were directly displaced by the Reform also lacked the capacity to resist, as their armies were "deliberately kept small" even relative to the headquarters force (Graff, 2003, pp. 230–31). Second, commissioners' losses were tolerable, as they still held control over the large headquarters army and were "by no means rendered powerless" (Peterson, 1979, p. 536).

Incomplete Reform. The rest of this paper documents that the Reform: (1) reversed the decline of central authority against the semi-autonomous armed actors (Figure 4, left column; Appendix Table B17, column 1), and (2) simultaneously increased mutinies and civilian uprisings (Figure 4, middle and right columns). One may wonder why the Tang rulers enacted the Reform, given its negative consequences, or why they did not introduce further measures to enhance efficiency or discipline afterward.

In hindsight, the Reform of 819 CE was "incomplete." It was the product of a weak state whose first-order concern was survival, with limited foresight

into local military dynamics. While we find that the emperors did achieve their primary objective of containing rebellions, they lacked the capacity to further reform the provincial military as local coercive power weakened. This specificity of Tang China does not, however, imply a rigid scope condition: incomplete or suboptimal reforms with unforeseen consequences are common throughout history and today (e.g., Chiovelli et al., 2023; Hellman, 1998). As Finkel & Gehlbach (2020) notes, reforms can be perilous when implemented by the local elites whose interests they seek to undermine. In our case, further efforts to curb local violence would likely have required cooperation from the "unruly" headquarters soldiers, whose uncertain compliance could have easily made matters worse.

What Is Tang Dynasty a Case of?

The Tang dynasty during its latter half provides a classic example of scenarios across space and time in which a weak center attempts to (re)assert *de facto* control over territories where it holds only *de jure* authority (e.g., Berman, 2010; Chen et al., 2025; Garfias, 2018; Migdal, 1988). Weakened by conflict, the Tang center faced significant challenges in exerting effective control over its territories.[20] This makes it an especially valuable case for testing theories of post-conflict state-building, where the central government seeks to restore its monopoly on violence in localities that saw its authority undermined by autonomous armed actors. These actors could hold nominal government titles conferred by the state, as demonstrated by various warlord-turned-governors in Afghanistan between 2001 and 2021 (Mukhopadhyay, 2014), or they could be non-state actors, such as the FDLR rebels in the Democratic Republic of the Congo that acted as stationary bandits before the 2009 Kimia II operation (Henn et al., 2025). The process of military centralization is expected to increase the central government's power over both types of actors, but may induce further downstream consequences. The Tang context is primarily relevant to the former category of actors. Appendix Section E discusses military reforms in the Byzantine Empire as a parallel case.

Data and Empirical Strategy

Data Sources

Commissioner-Controlled Armies. We undertake a comprehensive data collection effort to locate records of *commissioner-controlled armies in the outer prefectures* (hereafter "CCA prefectures"), between 769 and 818 CE. As the Reform transferred CCAs in outer prefectures from commissioner control to

the hands of central government agents, province-level variations in the number of outer prefectures with pre-Reform CCAs measure provinces' exposure to the Reform and underpins our empirical design.

For our main sources, we use the two official histories of the Tang dynasty and three imperially-commissioned works on history, politics and institutions that cover the Tang dynasty. We complement them with two additional works.[21] All of them are extensively used by historians to study the politics of Tang. See Appendix C.1 for detailed descriptions of these sources.

We identify CCAs using a keyword-based method. First, since CCAs were headed by commissioner-appointed commanders who were stationed in the outer prefectures, we obtain a list of titles usually conferred to those commanders from (Hino, 1980; Zhang, 2010).[22] Second, if a prefect was concurrently conferred military titles by the commissioner, the army in his prefecture would qualify as CCA (Zhang, 2010, p. 86).[23] We obtain from (Yen, 1969) a list of military titles that commissioners conferred.[24] We then perform a comprehensive search of these titles in our sources, and extract the locations at which the holders of these titles, and therefore their CCA, were stationed. We obtain 108 records with location information between 769 and 819 CE,[25] which map to 64 distinct outer prefectures with CCAs.

Figure 2 presents the province-level distribution of the number of CCA prefectures. Appendix Table B16 validates historians' understanding, mentioned in the Historical Background Section, that the CCAs were primarily a legacy of military activities during the An Lushan Rebellion (755 - 763 CE). One standard deviation in the number of battles during those eight years explains more than 50 % of a standard deviation in the CCAs, while most economic and geographic predictors are insignificant in the combined regression.[26]

Rebellions, Mutinies, Civilian Uprisings, and Battles. We construct a dataset on over 200 events of commissioner-led rebellions against the central government, mutinies of provincial headquarters armies against their commissioners, and civilian uprisings from a number of historical and modern scholarly sources.[27] Rebellions are commissioners' acts of open resistance to the central government via the use of provincial armies. Similarly, mutinies are acts of open resistance by rank-and-file officers and soldiers against their commissioners; by definition, these events took place in the headquarters prefectures where the commissioners resided, and involved only the headquarters army.[28]

Uprisings are armed conflicts initiated by civilians against the regime. For each event, we have information on its year, (prefecture-level) location of occurrence, and identity of the initiator(s). As explained in the Theory

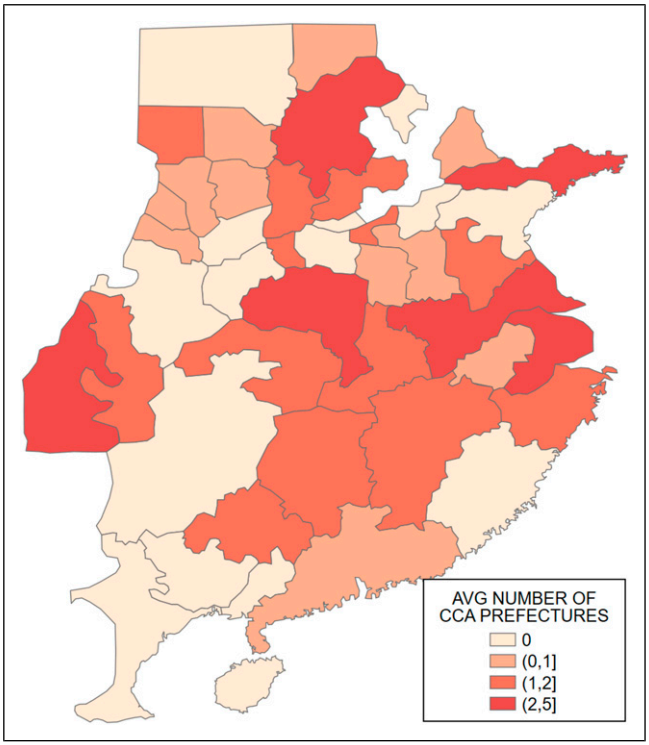


Figure 2. Distribution of CCA prefectures across provinces.
Note. This figure displays the number of outer prefectures with CCAs, averaged over the years 769 to 818 CE, at the province level. It is constructed using a base map of Tang in 820 CE, which we digitize from Tan (1996). The three autonomous provinces in the northeast (Youzhou, Chengde and Weibo) are excluded.

Section, holding grievances constant, an increase in uprising onset reflects weakened deterrence. We assume the likelihood of uprising to increase monotonically as coercive capacity declines, because participants in openly armed uprisings primarily respond to the state's perceived ability to repress effectively.

We also collect data on 420 battles that commissioners fought on behalf of the central government under imperial orders, including both defense against external enemies and repression of internal rebellions and uprisings.[29] For each battle, we know its year and location, participating commissioner and identity of the enemy. We consider a battle to be victorious for the central government if commissioner armies were recorded to defeat or repel their enemies, or acquire territory that was previously under enemy control.

Commissioners. We extract a list of 1,450 military commissioners of the Tang from (Wu, 2003), and obtain their biographical and officeholding information using the same sources as for the CCA records. For each commissioner, we have data on his hometown, ethnicity, mode of entry to officialdom (e.g., through the civil service exam (Wang & Yang, 2025; Wen et al., 2024), or military merit), whether his father was an official; the province, the first and last years of his commissioner appointments, and the reason for his departure (e.g., re-appointment to another position, death, or others).

Other Data. We collect the following geographical characteristics for each province and battle site: terrain ruggedness, river density, and suitability of rice and wheat.[30] We further compile data on population, natural disasters, and the total number of prefectures for each province, and control for them in our analysis.[31] Appendix Table B1 reports summary statistics for all variables used in the province-year and battle-level analyses.

Empirical Strategy

We apply a non-staggered difference-in-differences (DD) strategy to assess the impact of the Reform on the state's coercive capacity. According to Subsection "The Reform of 819 CE", provinces with a larger number of pre-Reform CCAs were affected more by the Reform, as their commissioners' command over provincial armies became weakened more. Thus, we measure provinces' exposure to the Reform using the average number of CCA prefectures over the years 769 to 818 CE.[32] We estimate the following equation at the province-year level:

$$Y_{pt} = \beta \text{NumCCAPref}_p \times \text{Reform}_t + \phi_p + \gamma_t + \epsilon_{pt} \quad (1)$$

Here, Y_{pt} denotes one of three outcomes related to military control and coercive capacity in province p and year t: commissioner-led rebellion against the center, soldier-led mutiny against the commissioner, and civilian uprising. For each outcome, Y_{pt} takes 1 if there are one or more instances and 0 otherwise. Results are consistent with both continuous measures (Appendix Table B6) and population-weighted outcomes (Appendix Table B8).

Our sample period is from 769 CE to 877 CE.[33] NumCCAPref_p is the average number of outer prefectures with records of CCAs between 769 and 818 CE in province p. Reform_t is a dummy equal to 1 if year t is 819 CE or later. Province fixed effects ϕ_p control for time-invariant provincial characteristics, and year fixed effects γ_t account for time shocks.

The coefficient β captures the causal effect of the Reform on the outcomes of interest. The key identifying assumption underlying the DD is parallel trend: provinces with fewer CCA prefectures provide an appropriate counterfactual for the trend in provinces with more CCAs, had the latter not been exposed to the Reform. We cluster standard errors by province.[34] Figure 5 verifies the absence of pre-trend through an event study exercise.

Figure 3 shows the raw outcome levels behind these estimates. For each decade it plots the share of province-years with at least one rebellion, mutiny, or civilian uprising, separately for provinces above and below the median number of CCA prefectures. Before 819 CE the high-exposure provinces rebelled more often than the low-exposure provinces and experienced mutinies and uprisings at similar or lower rates. After 819 CE the ordering reverses on all three outcomes, which is the pattern that the difference-in-differences estimates below quantify.

To further gauge the effect of the Reform on state's coercive capacity in the battlefield, we move our DD analysis to the battle-level, using a sample of battles that provinces fought on behalf of the Tang central government.[35] We estimate:

$$1(Win_{bt}) = \beta \text{NumCCAPref}_{bp} \times \text{Reform}_t + \phi_p + \gamma_\tau + \epsilon_{b\tau} \qquad (2)$$

Here, $1(Win_{bt})$ is a dummy equal to 1 if battle b in decade τ was a victory for the Tang. NumCCAPref_{bp} denotes the number of CCA prefectures in province p that participated in battle b. Decade fixed effects γ_τ account for time shocks common to all provinces. Reform_t and ϕ_p are the same as in equation (1).

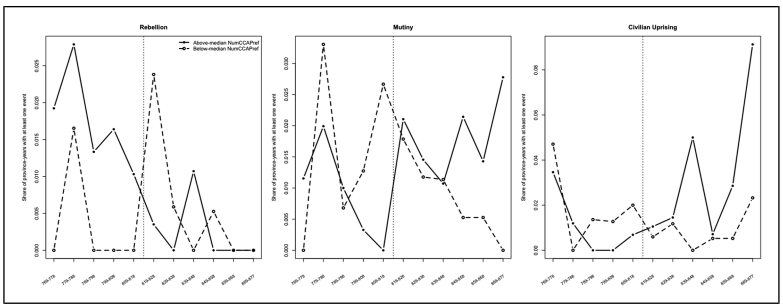


Figure 3. Raw trends in rebellion, mutiny, and civilian uprising.
Note. Each panel plots, by decade, the share of province-years with at least one event of the given type. Solid lines are provinces above the median number of CCA prefectures and dashed lines are provinces below the median. The dotted vertical line marks the Reform of 819 CE.

Reform and Coercive Capacity

Baseline Findings

Enhanced Central Control Over Commissioners. Before proceeding to conflict-related outcomes, we highlight a key result in column 1 of Appendix Table B17: the Reform strengthened the Tang rulers' control over the commissioners by increasing the frequency with which they departed their provinces due to centrally-mandated *reappointment* to other administrative offices. Thus, the Reform seemed to have achieved its primary goal of reasserting of central control over the provincial military powerholders.

Rebellions. Figure 4 reports the main findings from DD analyses at the province-year level. The left column shows that military centralization enhances the state's monopoly over violence by reducing commissioners' rebellion against the center. Control variables include terrain ruggedness, river density, logged population, distance to the capital, longitude, and latitude, all interacted with Reform_t as they are time-invariant. They also include province size measured by the number of prefectures, and the number of agriculture-affecting natural disasters as time-varying controls. The final specification adds region-year fixed effects to flexibly control for higher level time-varying unobservables.[36] Across specifications, for a one-standard-deviation increase in the number of CCA prefectures, the centralization reform led to a 0.47 to 0.72 percentage point decline in the probability of rebellion. As the average likelihood of rebellion in a given prefecture-year is .75%, the magnitudes identified here are sizable.

Mutinies. On the other hand, the Reform led to a decrease in coercive capacity (Figure 4, middle panel), as manifested by a deterioration in military discipline. Across specifications, the Reform led to a 1.14 to 1.66 percentage point increase in the probability of mutiny for a one-standard-deviation increase in the number of CCA prefectures. Given that the mean probability of mutiny is 1.3%, this negative effect on the discipline of the state's coercive apparatus is substantial.

Our mutiny result is best understood from a political opportunity perspective (Finer, 1974; Tilly, 1977). Recall that in the Tang dynasty, mutinies occurred at the provincial headquarters—where commissioners resided—and were led by headquarters soldiers. Before the Reform, CCAs in the outer prefectures acted as deterrence against potential mutinies. This is illustrated by the following example for the year 799 CE,[37] where an ally of the commissioner of Xuanwu province called on a CCA from an outer prefecture,

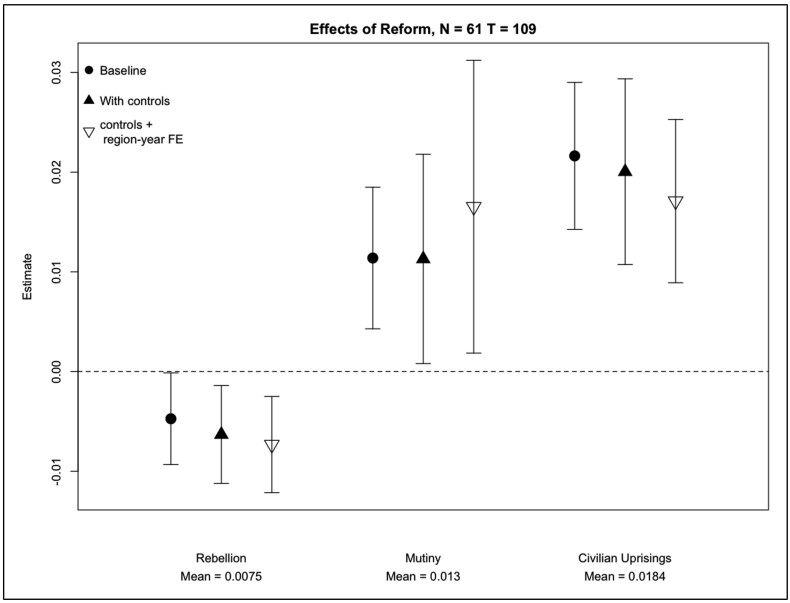


Figure 4. Effects of reform on the probability of rebellion, mutiny, and uprisings.
Note. Standard errors are clustered by province. Vertical bands represent 95% confidence intervals

Songzhou, then under the commissioner's authority, to quell a mutiny at the provincial headquarters.

> "The soldiers mutinied, killed the commissioner of Xuanwu province and his adjudicator, dismembered their bodies and ate them. Ju Wenzhen, the army-supervising commissioner, secretly summoned Liu Yizhun, the prefect of Songzhou [an outer prefecture of Xuanwu province] and a long-time general of Xuanwu province who was popular with the crowd; Yizhun led armies into Bianzhou [the headquarters prefecture of Xuanwu province] and quelled the chaos."

This instance highlights two features of the CCA's deterrence function, even though it had only arrived after the commissioner's death. First, the commissioner's camp had the authority to summon CCAs from outer prefectures in case of headquarters mutinies. Second, within the typical time frame of mutinies, which is several days, it is reasonable to expect that CCAs could ready themselves to stabilize the situation in the headquarters.

Additionally, this instance demonstrates the costly and risky nature of mutinies for their initiators, as reinforcements from the CCAs could arrive to crush the mutiny. The Reform eroded such deterrence by removing CCAs from the commissioner's control.

There was also pre-existing conflict of interest within the provincial military, which manifested itself in the form of soldier grievances, making the soldiers prone to mobilization in the first place (Gurr, 1993). New studies on civil-military relations (e.g., Grewal, 2019; Grewal, 2024) highlight military corporate interest to explain whether the military would support or challenge the status quo. As outlined in the Historical Background Section, headquarters soldiers shared the collective identity as armed but landless dwellers, and had a strong vested interest in remaining well-paid as bread earners. At the lower echelon of the military hierarchy, however, they were particularly vulnerable to payment delays and acutely aware of substandard military equipment. Grievances from a semi-hereditary soldiery reliant on state coffers drove what historians called "economic mutinies" (Qiu, 2018, ch. 6).

Building on the qualitative discussion above and the theoretical insights on military corporate interests mentioned in the Theory Section, we investigate how military centralization triggered mutinies. Specifically, we estimate the Reform's differential effects on mutiny in poorer versus wealthier provinces. We expect that the Reform's impact would be more pronounced in poorer provinces, where headquarters soldiers, facing greater financial vulnerability, were more likely to pursue their corporate interests against the commissioner. Drawing on (Zhang, 2010)'s classification of certain Tang provinces as "fiscal baskets" due to their rapid economic growth and substantial financial contributions since the 8th century, we designate these as wealthier, with the remaining provinces categorized as poorer. We then interact the Reform dummy with this wealth indicator. As expected, Appendix Figure A1 shows that the Reform had a significantly greater impact on mutiny in poorer provinces, where soldiers' financial grievances were more acute.

Historical records suggest that the central government treated mutinies with pragmatism. These incidents were typically confined to provincial militaries and rarely posed a broader threat to the regime. From the rulers' perspective, mutinies only became an existential threat when the province as a whole after mutiny was deemed openly defiant against imperial orders. There was thus political value of appeasing discontented soldiers. In many cases, mutineers expelled or even killed the commissioner but avoided assuming commissionership themselves, instead awaiting a new appointment of commissioner from the emperor. As historian (Qiu, 2018) argues (ch. 6), this practice of mutual restraint reflected a tacit arrangement: the mutineers

accepted a new commissioner from the emperor so that they would not be seen as openly defying imperial orders. Our data supports this view: of the 35 successful mutinies, only five leaders were later punished (four were killed in the province-level rebellions they subsequently staged against the emperor), while nine were formally recognized as commissioners. In the remaining 21 cases, the emperor appointed replacements whom the soldiers accepted. These patterns suggest that the rulers saw mutinies not as existential threats, but as manageable disruptions.

Uprisings. The right column of Figure 4 reports that the Reform also led to a large increase in civilian uprisings. Across specifications, the effect magnitude is 1.70 to 2.16 percentage points for a one-standard-deviation increase in the number of CCA prefectures, relative to an average rate of 1.84%. This sizable impact is an indication of worsened coercive capacity at the grassroots level, as centralization weakened the powerholders' ability to effectively deter potential rioters. We discuss alternative interpretations of this result in Subsection "Robustness Checks".

Parallel Pre-Trends. Figure 5 conducts a pre-trend test for our DD strategy. We aggregate the outcome variables to the decade level, and plot the coefficients for each decade across all three specifications. Estimates for pre-Reform decades are not indicative of statistically significant trends, lending further credence to a causal interpretation of our results.[38]

Robustness Checks

Our findings thus far indicate that the Reform strengthened the state's monopoly over violence, but simultaneously undermined its coercive capacity. These results are robust to an extensive battery of checks and tests, and are inconsistent with alternative explanations.

Alternative Coding of Key Variables. Results are qualitatively unchanged when using continuous outcome variables (Appendix Table B6); using a binary classification for treatment and control groups with cut-off at the median number of CCA prefectures (Appendix Table B5); and reconstructing the CCA variable using only official sources, which would reduce potential measurement errors (Appendix Table B11).[39]

While we have used the onset of uprisings as a measure of coercive capacity because its increase reflects weakened deterrence (see the next Subsection for a discussion), our results are also robust to a more direct measure of repression failure: the likelihood of *successful* uprisings.

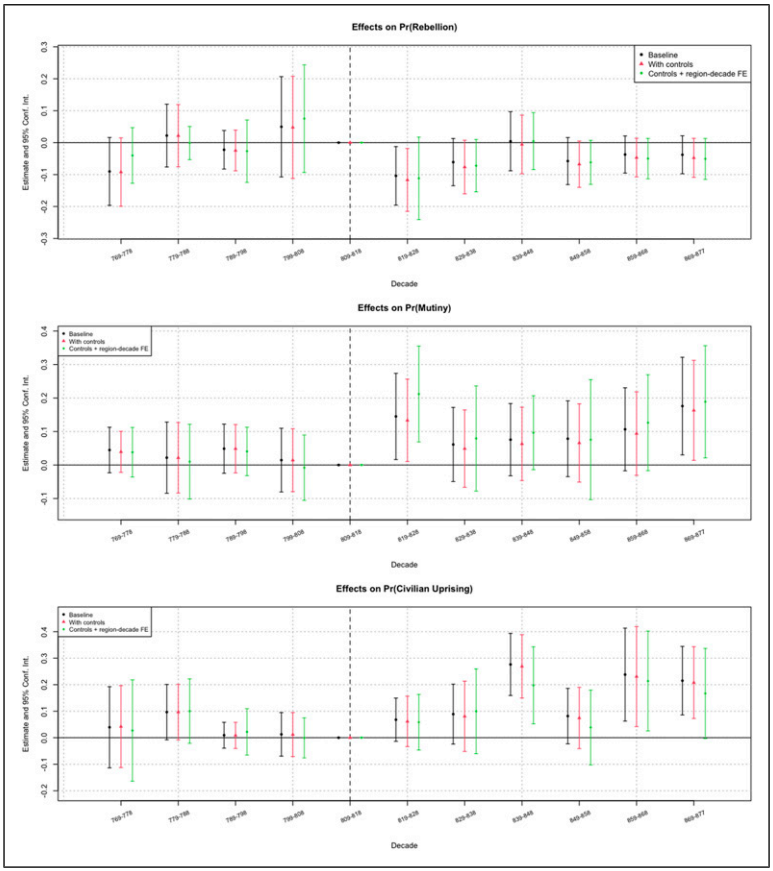


Figure 5. Tests for pre-trend.
Note. The top/middle/bottom subfigure presents decade-level lags-and-leads estimates for indicators of rebellion/mutiny/civilian uprising respectively. Standard errors are clustered by province. Vertical bands represent 95% confidence intervals.

Appendix Table B19 shows that the Reform increased the likelihood of uprising becoming successful.[40]

Reform Implementation. According to the imperial edict, the Reform was uniformly implemented with one exception: frontier garrisons (Dalby, 1979, p. 619).[41] It is thus plausible that the Reform's implementation was partial in frontier provinces—i.e., commissioners would still control armies in prefectures right at the border. We re-estimate our regressions, adding an

interaction term between a dummy for whether a province is situated at the imperial border and the Reform dummy. Our results are qualitatively unchanged (Appendix Table B9). Moreover, among the abundant historical records, only one province, Zelu, was specifically noted for incomplete implementation. Our results remain robust after excluding this province from the sample (Appendix Table B7).

To further check the degree of the Reform's implementation, we search for prefects holding concurrent military titles conferred by the commissioner in the *post*-Reform period, the exact same exercise we conducted to identify *pre*-Reform CCAs. This effort yields *zero* instances of such appointments post-Reform, indicating that the Reform had placed the CCAs under central authority as intended. As mentioned in Subsection "The Reform of 819 CE", the Reform was precipitated by the central government's decisive military victories over several notably rebellious provinces under Emperor Xianzong. Our results are also robust to including the interaction term between a dummy for provinces that were defeated by these campaigns and the Reform dummy (Appendix Table B10). Lastly, it is worth noting that incomplete implementation works against our favor—our results for rebellion would be biased towards zero if provinces under more recalcitrant commissioners received partial implementation.

Fiscal Policies. Though various changes occurred under Emperor Xianzong, potential compound treatment issues in our DD setup are largely avoided for two reasons. First, the Reform was implemented in the penultimate year of his reign and specifically targeted the CCAs. Second, we use a precise treatment variable constructed from specifically interacting the CCA prefectures *per se* with an indicator for years 819 CE and beyond. Nonetheless, we empirically address two fiscal policies from before and during Emperor Xianzong's reign that were not directly related to the CCAs. The first was the fiscal centralization of 809 CE and the second was the "Two Tax Policy" in 780 CE (for policy details, see Appendix C.2). Appendix Tables B2, B3 and B4 show that the results remain robust in these exercises.

Economic Development. As the military centralization reform specifically targeted the CCAs, it's unlikely that it would directly affect the economy. But potential economic changes might have been indirectly induced. Using population as a proxy for economic development, we show that provinces with greater reform exposure did not experience differential growth between 741 and approximately 980 CE (Appendix Table B14).[42] This non-result should mitigate the concern that our findings, especially on mutiny and

civilian uprisings, were driven by economic changes that the CCA-targeting Reform might have induced.

Commissioner Appointment and Characteristics. One may wonder whether the centralization reform led to the appointment of new kinds of commissioners to govern the provinces, and if so, whether our main findings are driven by the change in commissioner characteristics. While the Reform did increase the frequency of commissioner reappointments (column 1), findings in columns 2 to 7 of Appendix Table B17 suggest no significant commissioner differences in terms of ethnicity, mode of entry into officialdom, and father's political experience. Furthermore, causal mediation analysis suggests that the increase in commissioner reappointments unlikely explains the total effects of the Reform on rebellion, mutiny, and uprising (see Appendix C.3 and Figure A3 for details).

Reduced Army Channel. Finally, (Peterson, 1979) notes that Emperor Muzong issued a secret edict (in effect 820–822 CE) ordering provincial armies not to replenish soldiers lost through attrition, raising the concern that our findings could reflect troop reductions during the Muzong window rather than the Reform. Appendix Table B21 shows the centralization effects on all three outcomes are robust to controlling for Battles × Muzong and Lost Battles × Muzong. Because these controls are post-treatment, we additionally estimate the average controlled direct effect (ACDE) of the Reform via sequential *g*-estimation (Acharya et al., 2016), treating both Muzong interactions jointly as mediators. The ACDE estimates are similar to the total effects (Appendix Figure A9). Appendix Figure A10 reports sensitivity analyses on the sequential unconfoundedness assumption: an unobserved confounder would have to induce an implausibly large correlation between the mediators and the outcome errors to overturn the ACDE estimates.[43]

How Military Centralization Induced Civilian Uprisings

Reduced Deterrence. Recall our argument in the Theory Section that, by amplifying communication and coordination problems among the commissioner's military units, the Reform reduces the military's coercive capacity by undermining their ability to: (a) deter potential rioters, and (b) contain existing uprisings. To disentangle the two possibilities, we decompose uprisings into two types: initial uprisings, which are newly-launched; and subsequent uprisings, which follow from the spread of initial uprisings from other places. Appendix Table B15 shows that both types increased, suggesting that both the deterrence and the containment mechanisms are at work.[44]

The deterrence mechanism presumes that civilians were aware of the power shake-up within the provincial military, and performed rational calculations of repression probability using such knowledge. One may worry that uprisings in historical China were driven not by such calculations, but by grievances stemming from economic downturns or mistreatment by government officials. While grievance had always been a factor for civilian uprisings throughout Chinese history, it is worth emphasizing that the surviving records of uprisings from medieval times are not as extensive as records of the early modern period, and the uprisings tend to be sizable ones that drew prefectural or provincial attention. Thus, it is highly plausible that these activities were led by individuals who were knowledgeable and sophisticated enough to calculate the expected gains from uprising. Moreover, since provincial armies were locally recruited, it is also possible that the soldiers shared institutional knowledge of the military with their peer civilians.[45] Since the civilians who led these uprisings plausibly understood the shift in military control, the decomposition in Appendix Table B15 indicates that the centralization Reform reduced provincial armies' coercive capacity by lowering their ability to deter new uprisings as well as to repress existing ones.

Alternative Explanations. We now consider two alternative explanations for our civilian uprisings result. One is that mutinies "distracted" soldiers from deterring uprisings. Another interpretation is that the unruly headquarters soldiers were detrimental to the local society, essentially acting as sources of local intimidation and thereby causing grievances among civilians. We present evidence against these two interpretations via sequential *g*-estimation of controlled direct effect (ACDE) of the Reform following (Acharya et al., 2016).

For the first interpretation, we use civilian uprising as the outcome, with mutiny as a "mediator", in our DD design. The top panel of Appendix Figure A2 shows that the ACDE of Reform on uprising is almost identical to the total effect of Reform, suggesting that most of the impact was unlikely mediated by the mutiny-distraction channel. As a robustness check, we further restrict our focus from civilian uprisings in the entire province to ones that happened in the provincial headquarters, where soldier-led mutinies took place. Again, the ACDE of Reform is very close to the total effect (middle panel). The bottom panel reports sensitivity analyses for the sequential unconfoundedness assumption for both tests, suggesting that the potential unobserved confounder affecting both mediator and outcome would have to be exceedingly large to render the ACDE indistinguishable from zero.[46]

For the second alternative explanation, we examine civilian uprisings in the headquarters and outer prefectures separately. If soldiers exacerbated

civilian grievance (hence uprising), then we should see centralization causing an increase in uprisings in the headquarters, where the unruly soldiers were located, but not in the outer prefectures. Appendix Table B13 indicates that the effect is similarly pronounced in both locations.

Another interpretation is that prefects, as representatives of the central government, had weaker ties to the communities they governed. When military centralization transferred the control over outer prefecture CCAs from commissioners to prefects, the shift might have reduced the effectiveness of prefecture-level military leadership, weakening prefecture armies' ability to deter and thus leading to more civilian uprisings. This interpretation is more likely to be valid if the effect of the Reform on civilian uprisings is stronger in outer prefectures than in provincial headquarters. However, Appendix Table B13 shows that increases in civilian uprisings were comparable for outer prefectures vs. headquarters.[47]

A further question is whether the rise in non-headquarter uprisings concentrated in the specific prefectures whose armies the Reform transferred to central control. We constructed a prefecture-year panel of non-headquarter prefectures, distinguishing those that had a commissioner-controlled army before 819 CE ("CCA prefectures") from those that did not. Appendix Figure A7 shows that, within the same province-year, CCA prefectures experienced a substantially larger post-Reform increase in the probability of civilian uprising than non-CCA prefectures, with the pattern surviving the inclusion of province-year fixed effects that absorb all time-varying province-level shocks. The increase emerged gradually across post-Reform decades rather than immediately after 819, which is inconsistent with an interpretation centered on demoted-commander backlash (Appendix Figure A8).

Battle-Level Results

Finally, we explore how the Reform's negative impact on coercive capacity manifested itself in provincial armies' performance in the battlefield. We move our analysis to a sample of 420 battles, in which commissioners were mandated by the central government to take part and fight on behalf of the dynasty. We estimate equation (2), and results are displayed in Table 1. Control variables include whether a central army from the Tang capital was present at the battle. They also include battle-site characteristics: longitude, latitude, terrain ruggedness, distance to the Tang capital, distance to the participating army's provincial headquarters, suitability indices of rice, millet and wheat, all interacted with Reform_t. Enemy type fixed effects distinguish between foreign enemies, internal Chinese rebels, and ethnic minorities residing in Tang territory.

Table 1. Effect of Reform on Battle Success

	Indicator of battle victory for Tang					
	(1)	(2)	(3)	(4)	(5)	(6)
NumCCAPref × reform	−0.125** (0.047)	−0.185*** (0.034)	−0.181*** (0.038)	0.099 (0.194)	0.054 (0.116)	0.071 (0.112)
NumCCAPref × reform × CCA avg. distance				−0.106 (0.098)	−0.109* (0.063)	−0.121* (0.064)
Controls	No	Yes	Yes	No	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes	Yes	Yes
Enemy type FE	No	No	Yes	No	No	Yes
Observations	434	434	434	350	350	350
R-squared	0.25	0.33	0.33	0.23	0.33	0.33

Note. This table shows the effect of Reform on battle victory. CCA Avg. Distance is the average distance between CCA prefectures and the provincial headquarters. Robust standard errors in parentheses are clustered at province level. ***, **, * denote significance at 1%, 5%, 10% levels.

All three specifications across columns 1–3 indicate that military centralization lowered the probability of winning battles. For a one-standard-deviation rise in the number of CCA prefectures, the Reform led to a 12.5 to 16.6 percentage point decline in the probability of winning for the Tang armies. Importantly, these results are not driven by sample selection because the Reform did not cause higher participation for provinces with a higher number of CCA prefectures (Appendix Table B12).

Evidence for the Communication and Coordination Mechanisms. We present evidence for communication and coordination problems as a key mechanism in our theory. For quantitative evidence, we build on equation (2) to further interact the main treatment variable ($\text{NumCCAPref}_{bp} \times \text{Reform}_t$) with the average distance (in 100 kilometers) from each pre-Reform CCA prefecture to the provincial headquarters, calculated for each province, while controlling for the constituent terms in the three-way interaction specification. As communication and coordination challenges should increase in the distance from military units to the command headquarters, they should become more severe for provinces with longer average distances between the pre-Reform CCAs and the commissioner after the Reform induced incoherence in the command structure. Columns 4–6 of Table 1 provide support for this expectation: the triple-interaction coefficients are negative across specifications

and statistically significant in the two controlled specifications. The negative impact of the Reform on Tang military performance in the battlefield was stronger for provinces with longer distances.[48]

In addition, Appendix C.4 presents a detailed case drawn from primary historical sources demonstrating how coordination failures undermined Tang military effectiveness after the Reform. In the Pang Xun Uprising (868–869 CE), a disjointed and incoherent military response, caused by multiple failures of different prefectural armies to coordinate with the provincial command, led to a series of devastating defeats and the fall of the provincial headquarters. This example underscores how the Reform, while strengthening central authority, sowed dysfunction between provincial commissioners and their subordinates.

Conclusion

States are built around their effective use of coercion. The presence of local powerholders with substantial military capacities presents formidable challenges to rulers who strive to consolidate their monopoly over violence. While state-building through military centralization may emerge as a means to constrain the powerholders, it entails a restructuring of the state's coercive apparatus as a whole, and would inevitably affect the state's ability to exercise its coercive capacity.

Our paper theorizes that military centralization can enhance monopoly over violence but simultaneously erode the state's coercive capacity. We empirically explore this argument in the context of Tang China, which had a weak center and strong local military powerholders. We document that military centralization reduced rebellions initiated by local military powerholders against the center and increased the central government's control over these powerholders via imperially-mandated departures. Meanwhile, it undermined the state's ability to enforce internal political order and to counter threats to the regime, as reflected by an increase in soldier-led mutinies, civilian uprisings, and a decline in the probability of winning battles.

Our findings connect with qualitative studies of historical and contemporary regimes across the world, and with military dynamics between rulers and national armies. In addition to the Byzantine case, the experiences of Vietnam, Iraq (Talmadge, 2015) and the Philippines (Greitens, 2016) also speak to how successful centralization can weaken coercive capacity in other respects. With ongoing challenges from autonomous armed actors in many societies (Berman, 2010), lessons from medieval China could draw scholarly and policy attention to the paradoxical consequences that arise when state-building encounters the guardianship dilemma.

Author's Note

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The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement

Replication materials, including data and code, can be found at (Chen & Wang, 2026).

Supplemental Material

Supplemental material for this article is available online.

Notes

1. Prefectures are administrative subdivisions of provinces. Each province has one headquarter prefecture and multiple outer prefectures. The commissioner of the province resides in the headquarter prefecture.

2. Commissioners could lawfully command these "outer prefecture" armies indirectly through the prefects when they participate in centrally-mandated military missions. See discussion in Subsection "The Reform of 819 CE".
3. See Chen et al. (2025); Garfias & Sellars (2026); Garfias & Sellars (2021); Lee & Zhang (2017); Wang (2022) for extractive capacity; and Guardado (2018); Gulzar & Pasquale (2017); Toral (2024) for administrative capacity Hanson & Sigman (2021) discusses how the three dimensions underpin each other.
4. We use this term interchangeably with "military centralization" throughout the paper.
5. The powerholder likely prefers individuals who have shared work experience or hometown ties with him, and are therefore more familiar with his work style, character, and methods of communication.
6. Military coordination can also be harmed by frequent rotation of commanders which seeks to avoid the formation of commander-soldier ties (Gaub, 2013; Nassif, 2015a). Pollack (2002, pp. 358–424) points out that reduced training impairs the army's ability to coordinate the efforts of artillery, armor and infantry. In Appendix C.4, we discuss a case from the post-Reform period to illustrate how communication and coordination problems could impair the operations of the Tang provincial military.
7. This mechanism implicitly assumes elite non-monolithicity, which allows for the impact of military centralization to differ across various segments of the military organization.
8. Personnel policies such as recruitment and promotion are core instruments of cooptation and control in authoritarian hierarchies (Svolik, 2012). By rotating subordinates through positions and shaping their career trajectories, they also erode the "local embeddedness" that would otherwise bind subordinates to the powerholder (Hassan, 2020, p. 38). Relatedly, Chen (2024) finds that military centralization is associated with outward military aggression.
9. See Appendix C.4 for qualitative evidence to illustrate communication and coordination between the headquarters army and prefect-led armies.
10. These independent and hereditary dukedoms in Hebei recognized the Tang emperors ceremonially, and the emperors rarely attempted to interfere with any political affair inside them. We thus exclude the three Hebei provinces from our analysis.
11. In addition to provincial armies, the emperors had at their disposal the Imperial Palace Army based in the capital city. This central army formed the backbone of central military power and contributed to the Tang Dynasty's survival (Graff, 2003, p. 234).
12. This is the seat, or administrative center, of the provincial government.
13. All recorded mutinies against commissioners originated from the headquarters army. Soldiers in the outer-prefecture CCAs would have to travel to the headquarter prefecture to mutiny, which was much more difficult.

14. This transfer of military command did not reduce the the commissioner's military force, as he could still "muster and lead the army of his whole province, provided he were under imperial orders to do so" (Dalby, 1979, p. 619). That is, the Reform sought to constrain the commissioners' potential to cause trouble while preserving their ability to command local armies for imperial missions.
15. To verify this, we utilize the fact that *commissioner*-appointed prefects carried the title of "acting prefects" (Yan, 2023, pp. 158–59). Using a list of all recorded Tang prefects from Yu (2001), we find that only 0.98% of all prefects in our period of study were "acting prefects"—that is, 99.02% were centrally-appointed.
16. A proposal by senior official Wu Chongyin, which became the Reform, argued that a commissioner's rebelliousness stemmed from his "ability to mobilize the entire armed force of his province" (Dalby, 1979, p. 619; Peterson, 1979, pp. 535–536). Wu suggested returning the military functions of commissioner-appointed commanders in the outer prefectures to the prefects. We find supporting evidence for his claim.
17. Pre-Reform CCAs are indeed identified via a list of commissioner-conferred military titles (Subsection "Data Sources").
18. Prefects had fewer incentives to defend the commissioner because it was outside their sphere of responsibility, involved high personal risk, and rewards were uncertain compared to the pre-Reform period.
19. The exclusion of the autonomous northeastern provinces, where the center lacked the relative strength to impose reform, illustrates the converse. Our results are robust to controlling for Emperor Xianzong's pre-Reform campaign victories against the provinces (Appendix Table B10).
20. While historical comparisons often depict China as a strong and centralized state relative to Western Europe, the mid-to-late Tang dynasty represents a notable exception within Chinese history.
21. Our results are robust to the exclusion of these two non-official sources (Appendix Table B11).
22. The titles include: *zhen (e) jiang* (garrison general); *zhen (e) shi* (garrison commissioner); *zhen'e (duzhi) bingmashi* (garrison commander); and *(tong) jiedu fushi* ((co-)vice military commissioner).
23. We verify that concurrent titles as such disappear in the post-Reform period.
24. The titles include: *duzhi bingmashi* (supreme commander); *zuoxiang/youxiang/houyuan deng bingmashi* (commander of the left/right/back armies); *yuhou, du-yuhou* (provost marshal); *yaya, du-yaya* (general). Of these 16 titles, 15 appear in our data.
25. They constitute 71.3% of all records.
26. Our identifying assumption in Subsection "Empirical Strategy" is parallel-trend and does not require the random assignment of the CCAs.

27. Sources: *Comprehensive Mirror*, the two *Books of Tang*, Wang (1969); Zhang (1979) and Zhang (2010).
28. That mutinies in our context occurred only in the "capitals" of territorial units is consistent with cross-national data in (Dwyer, 2018), which suggest that modern mutinies predominantly happen in national capitals.
29. Sources: the *Catalogue of Historical Wars* (*Zhongguo Lidai Zhanzheng Nianbiao*, compiled by China's Military History Editorial Committee), complemented by the *Comprehensive Mirror*.
30. Sources: USGS, CHGIS, and FAO.
31. Sources: Liang (1980); the *Collection of Meteorological Records in China over Three Thousand Years (Zhongguo Sanqiannian Qixiang Jilu Zongji)*, volume 1, edited by De'er Zhang and compiled by the China Meteorological Administration; Guo (2012).
32. Our baseline findings are robust to an alternative, binary classification of treatment and control groups, using the median number of CCA prefectures as the cut-off (Appendix Table B5).
33. We choose 769 CE as the starting year because the political unrest following the rebellions led by An Lushan and Pugu Huai'en had largely subsided by this time. We choose 877 CE as the end year because Huang Chao, a rebel who "emerged as the dominant bandit leader in 878 [CE]," would deal irreparable damage to the Tang imperial power over the next few years, during and after which provincial commissioners would become independent warlords (Graff, 2003, pp. 242–43).
34. Results for all three outcomes remain robust when replacing province-clustered standard errors with Conley spatial standard errors, which allow for arbitrary correlation across geographically proximate provinces (Conley, 1999). See Appendix Figure A4 for estimates across distance cutoffs of 100, 300, and 500 km.
35. We find no evidence of selection on the sample of battles: the Reform did not cause higher participation for provinces with a higher number of CCA prefectures (Appendix Table B12).
36. Region (*dao*) is a term used in the Tang government to group large numbers of geographically adjacent provinces based on common characteristics in economy and society. Tang did not have permanent government offices at this higher administrative level.
37. *Comprehensive Mirrors* (volume 235).
38. A year-level event study restricted to a 10-year window around the Reform (809–828 CE) likewise shows no pre-trend, though post-Reform estimates are less precise given the much smaller sample (Appendix Figure A6). Re-estimating equation (1) on this 10-year subsample yields similar effects for rebellion and mutiny (Appendix Figure A5); the absence of an immediate effect on civilian

uprisings is consistent with our framework, in which weakened military discipline precedes the erosion of deterrence.

39. The more official sources have a relatively unified standard on recording local events, which tended to be major instances. This way our alternative coding also alleviates the concern that the commissioners whose interests were undermined by the Reform would over-report local uprisings to the center as a signal of the Reform's negative impact.
40. We also probe Tang military failures against domestic enemies in *battle-level* regressions using the framework introduced in Subsection "Battle-Level Results". Appendix Table B20 shows that the negative impact of the Reform on military effectiveness in the battlefield is similar against both domestic and foreign enemies.
41. The edict of 819 CE states: "In subordinate prefectures under provincial commissioners, except the headquarters prefecture, troops stationed in separate garrisons shall all be placed under the prefects' authority. […] Fortresses specially established in frontier areas adjoining non-Chinese territories and outside prefectural jurisdiction are exempt." *Tang Huiyao* (Vol. 78; our translation). The original Chinese reads: 诸道节度使, 团练, 都防御, 经略等使所管支郡, 除本军州外, 别置镇遏守捉兵马者, 并合属刺史等。……其有边于溪洞, 接连蕃蛮之处, 特建城镇, 不关州郡者, 即不在此限. Dalby notes that the Reform probably had no effect in the autonomous northeastern provinces, which are excluded from our sample (Section 2.2).
42. Household data for the Tang and Song dynasties are available for the years 742 and approximately 980 CE, respectively, from (Liang, 1980).
43. A further check confirms that the Reform did not increase battle activity or lost-battle counts in high-exposure provinces during the Muzong window itself (results available upon request), ruling out a mechanical link between Reform exposure and the Muzong-window mediators.
44. Consistent results with spatial standard errors (Conley, 1999) are available upon request.
45. Similarly, Egyptian protesters acquired insider knowledge of the army through their friends and contacts in the military (Cebul & Grewal, 2022).
46. This is particularly true for the first test. For the second test, note that a relatively moderate ρ in the *negative* direction would affect our results. However, potential unobserved factors that would jointly affect mutiny and uprising were most likely province-time specific fiscal or financial shocks, which should have affected soldier and civilian grievance in the *same* direction (i.e., positive ρ).
47. We further confirm the lack of differentiation by constructing a variable that takes the *difference* between the number of headquarters vs. outer prefecture civilian uprisings and regressing this new outcome on the same DD specification. The Reform did not appear to statistically increase or decrease this variable.

48. This exercise is empirically demanding because it substantially reduces our degrees of freedom: provinces without CCAs would not have an average distance metric and thus are dropped from the analysis.

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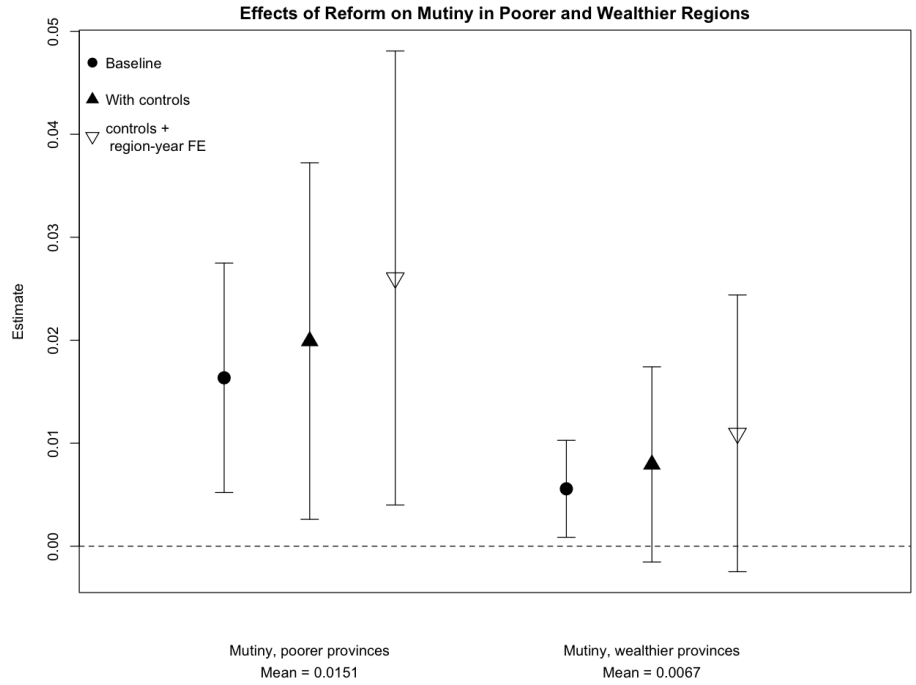
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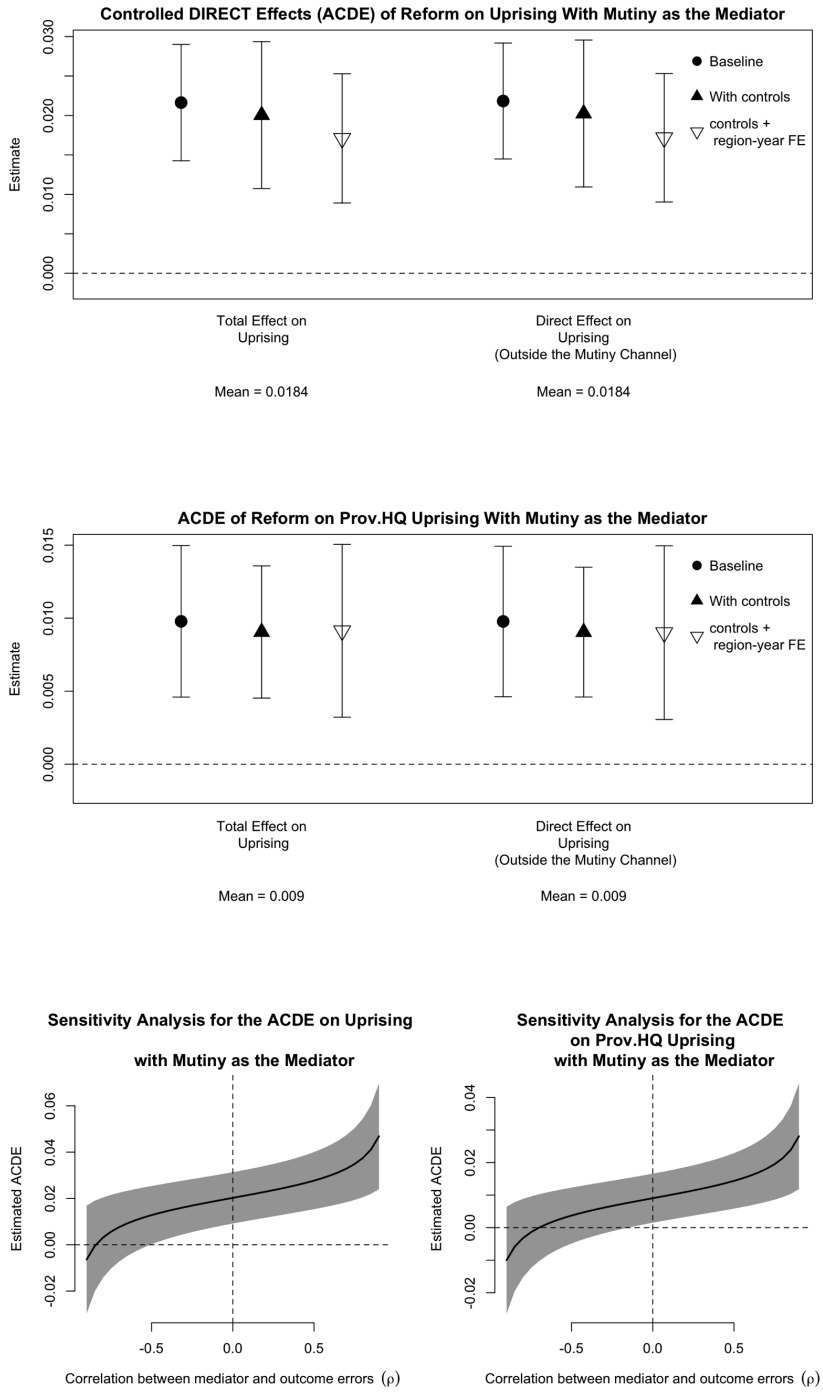
A Additional Figures

Figure A1. Effects of Reform on the Probability of Mutiny, Poorer vs. Wealthier Provinces



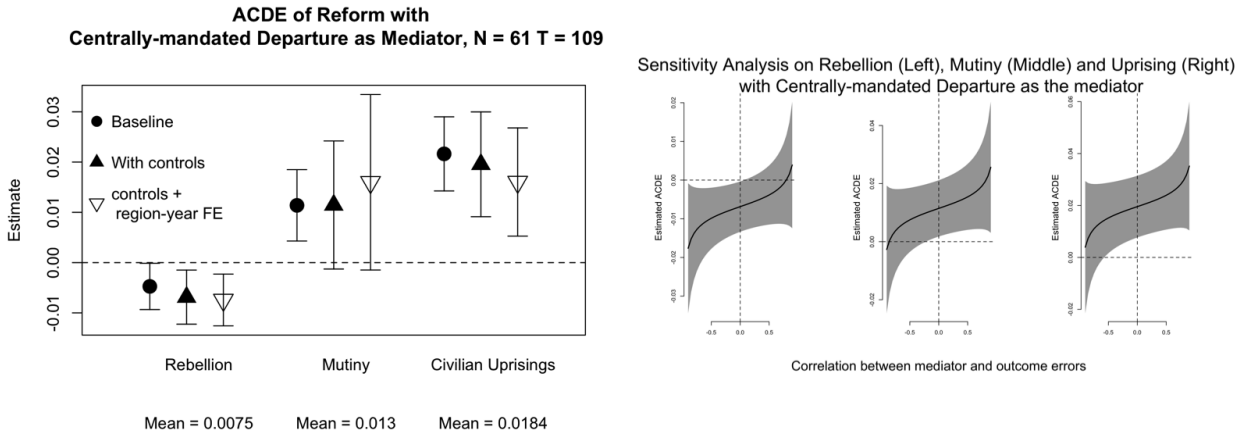
Notes: Standard errors are clustered by province. Vertical bands represent 95% confidence intervals. The left panel displays the coefficients on NumCCAPref × Reform, which capture the effect of the Reform on poorer provinces. The right panel displays the sum of the coefficients on NumCCAPref × Reform and the triple-interaction term, which capture the effect of the Reform on wealthier provinces, and computes the standard error associated with that sum.

Figure A2. Effects of Reform on Uprising, with Mutiny as Mediator



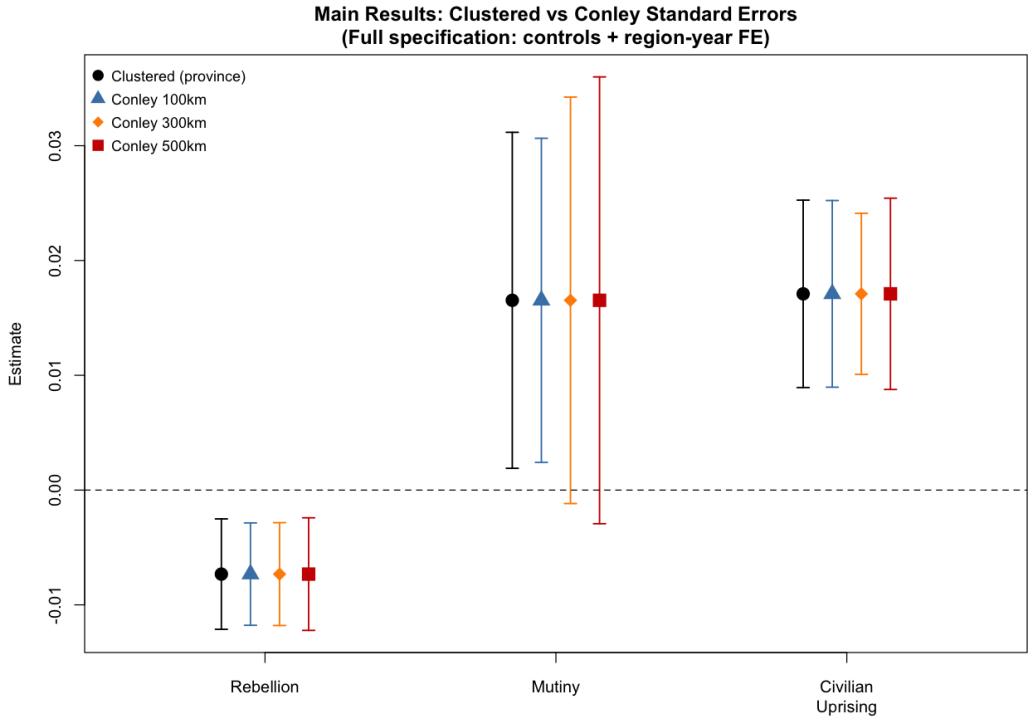
Notes: The top and middle panels show the controlled direct effects of reform on the probability of overall uprising and uprising at headquarters prefectures, compared with total effects. The bottom panel shows the corresponding sensitivity analyses in the causal mediation framework. Standard errors are clustered by province. Vertical bands represent 95% confidence intervals.

Figure A3. Effects of Reform, with Centrally-Mandated Departure as Mediator



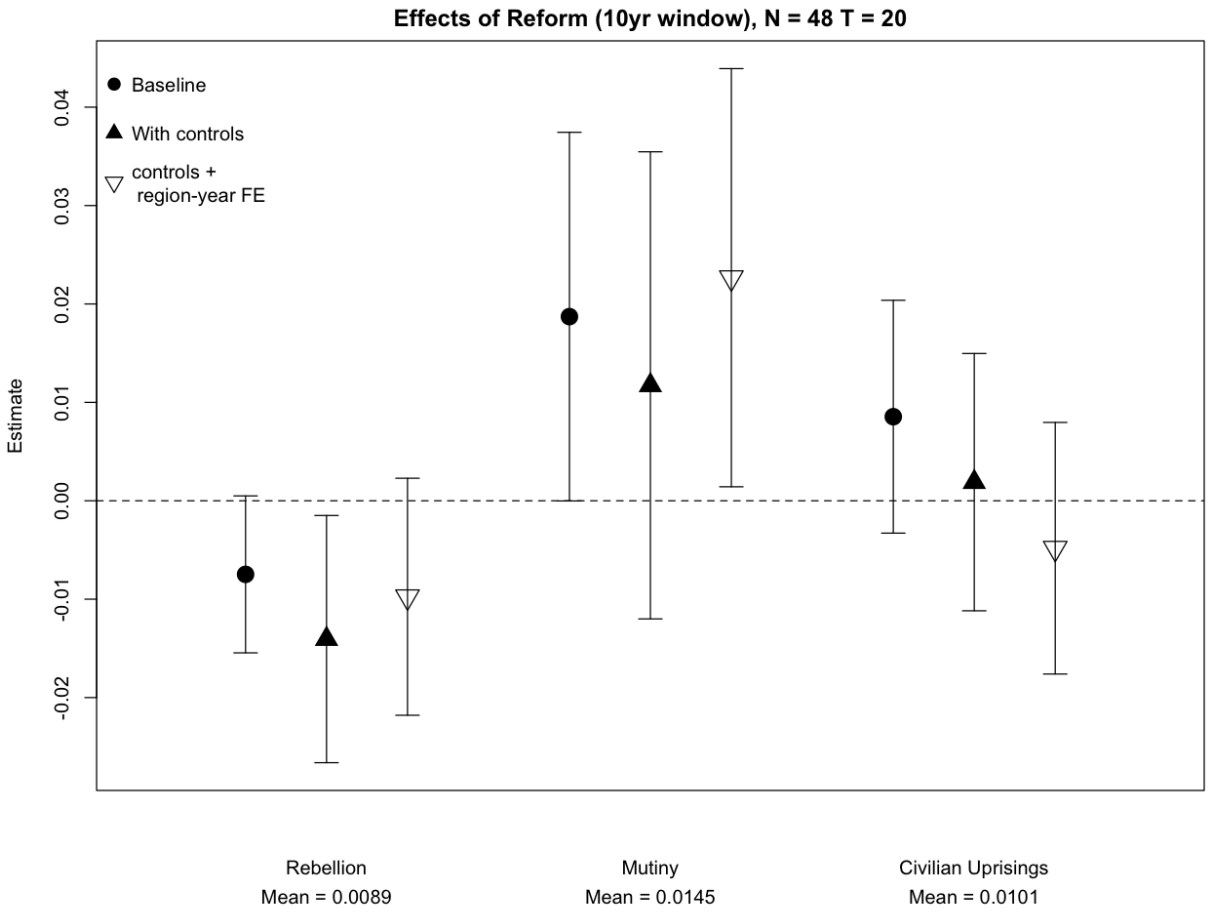
Notes: This figure shows the controlled direct effects of reform on the probability of rebellion, mutiny, and uprising, with centrally-mandated departure as the mediator. The ACDE (sensitivity analysis) is on the left (right). Standard errors are clustered by province. Vertical bands represent 95% confidence intervals.

Figure A4. Effect of Reform under Conley Spatial Standard Errors



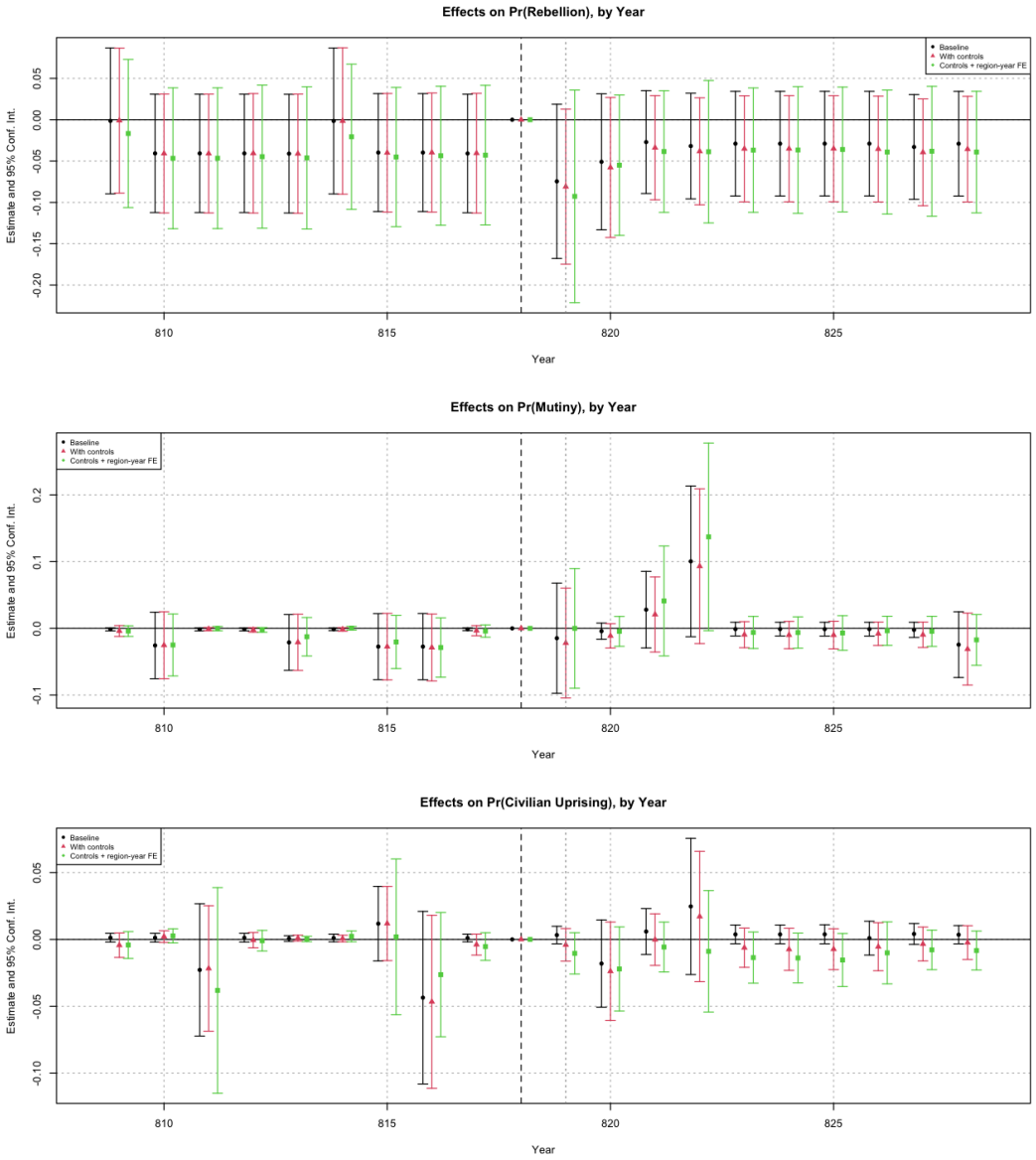
Notes: This figure displays the estimated effect of the Reform on the probability of rebellion, mutiny, and civilian uprising, using the fully-controlled specification (controls + region-year fixed effects). Each outcome is shown with four confidence intervals: province-clustered standard errors (the baseline used in the main analysis), and Conley (Conley 1999) spatial standard errors computed with cutoff distances of 100 km, 300 km, and 500 km. Vertical bands represent 95% confidence intervals.

Figure A5. Effects of Reform, 10-Year Pre/Post Subsample



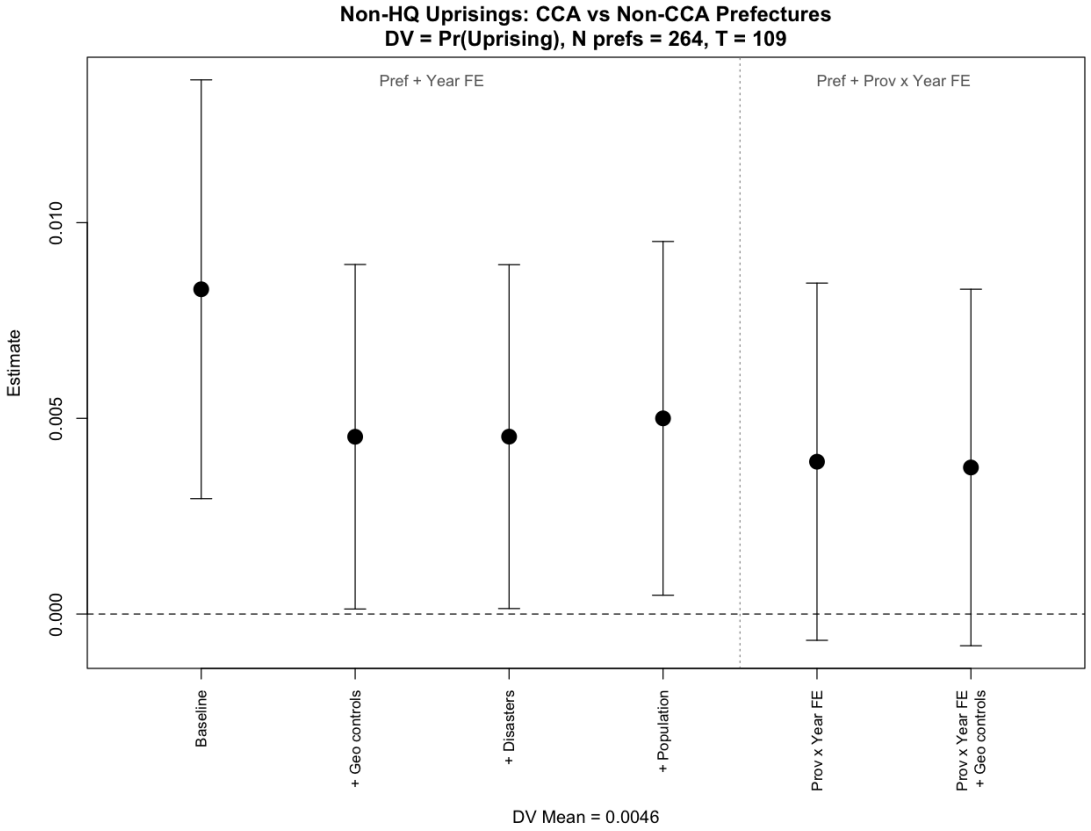
Notes: This figure replicates Figure 4 on a 10-year window symmetric around the Reform (809–828 CE). Standard errors are clustered by province. Vertical bands represent 95% confidence intervals.

Figure A6. Event Study by Year, 10-Year Pre/Post Subsample



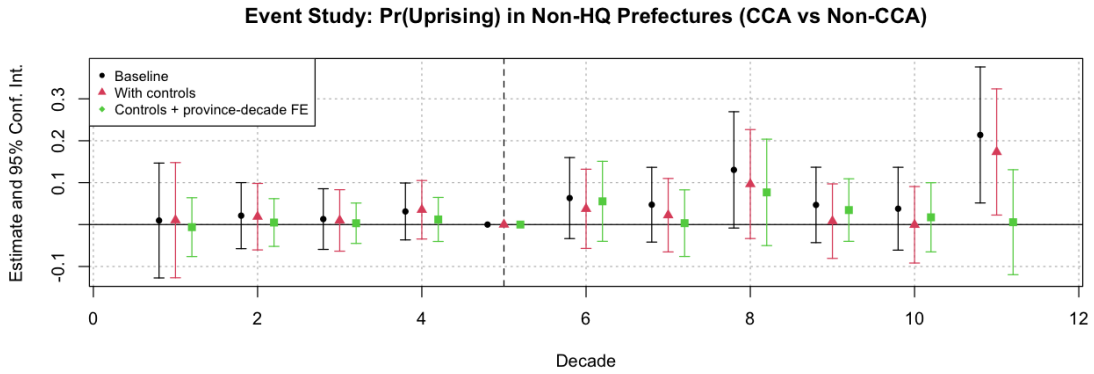
Notes: This figure presents a year-level event study of the Reform's effects on the probability of rebellion (top), mutiny (middle), and civilian uprising (bottom), restricted to a 10-year window symmetric around the Reform (809–828 CE). Each coefficient is the interaction between *NumCCAPref* and an indicator for the corresponding year, with year 818 (the last pre-Reform year) as the omitted reference. The three colors correspond to three specifications: baseline (black), with controls (red), and controls + region-year fixed effects (green). Standard errors are clustered by province; vertical bands represent 95% confidence intervals. The vertical dashed line marks 819 CE, the year of the Reform.

Figure A7. Effect of Reform on Uprisings in Non-Headquarter Prefectures, by CCA Status



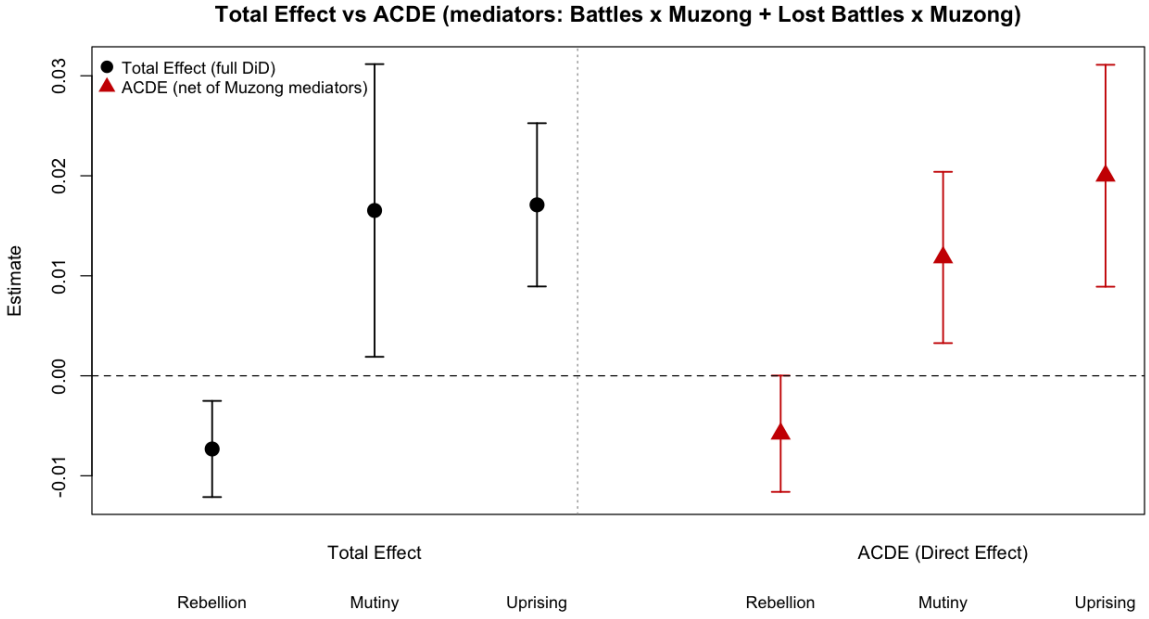
Notes: This figure presents prefecture-year level estimates of the effect of the Reform on the probability of a civilian uprising, restricted to non-headquarter prefectures. The unit-level treatment is a binary indicator equal to 1 if the prefecture had a CCA before the Reform, interacted with the post-Reform dummy. Specifications 1–4 include prefecture and year fixed effects with progressively richer controls; specifications 5–6 add province-year fixed effects, which absorb all time-varying province-level shocks. Standard errors are clustered by province. Vertical bands represent 95% confidence intervals.

Figure A8. Event Study: Non-Headquarter Uprisings in CCA vs. Non-CCA Prefectures



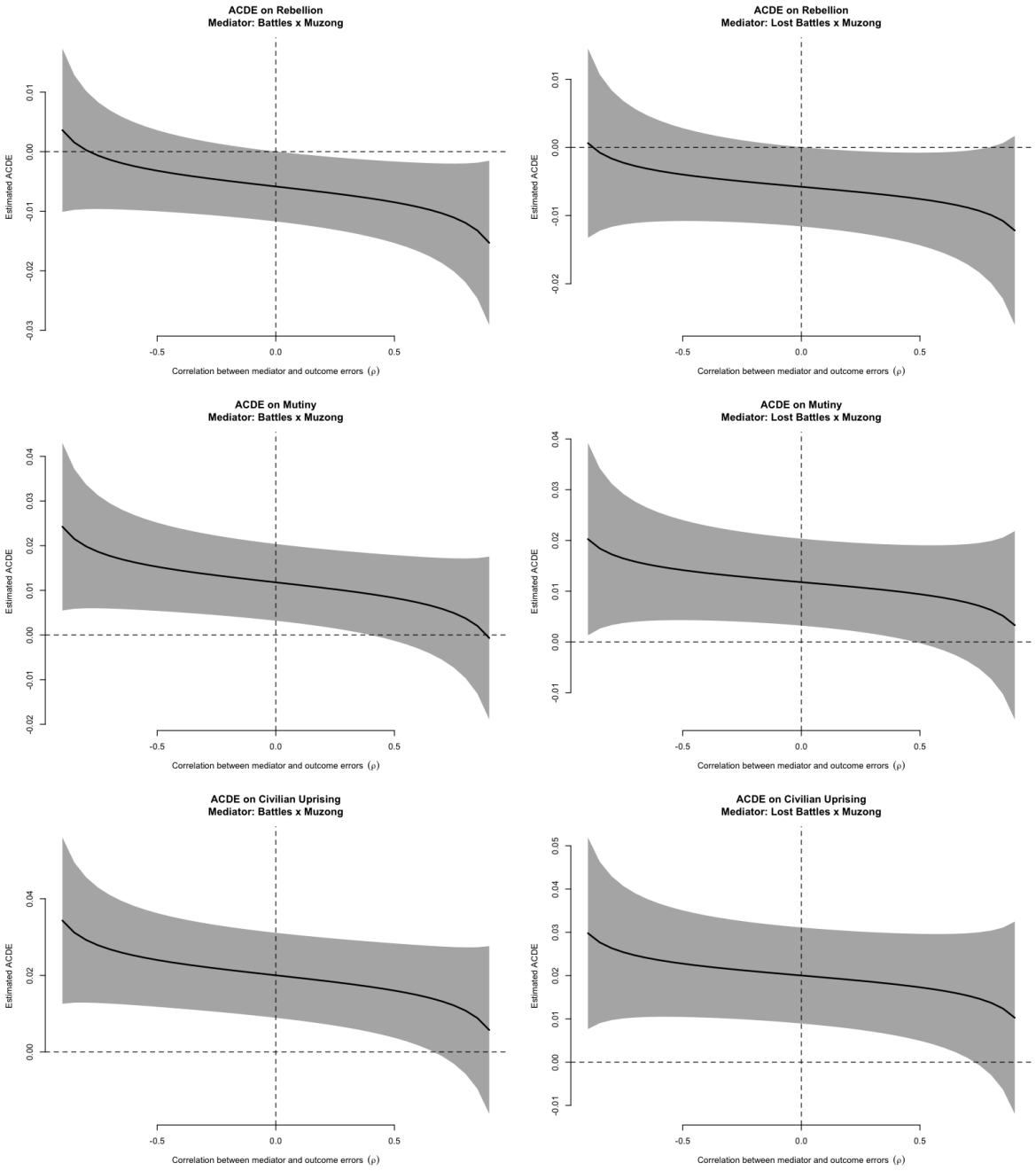
Notes: This figure presents decade-level event-study estimates of the interaction between an indicator for a non-headquarter prefecture having a CCA before the Reform and decade dummies. Decade 5 (809–818 CE, the last pre-Reform decade) is the omitted reference. The three specifications correspond to: baseline with prefecture and decade fixed effects (black); with prefecture-level controls (red); and with province-decade fixed effects (green). Standard errors are clustered by province. Vertical bands represent 95% confidence intervals.

Figure A9. Total Effect vs. Average Controlled Direct Effect (ACDE), Muzong-Window Mediators



Notes: Each pair compares the total effect of the Reform (left, black) to the average controlled direct effect (ACDE, right, red), with *Battles* × *Muzong* and *Lost Battles* × *Muzong* entered jointly as post-treatment mediators. ACDE estimated via sequential *g*-estimation (Acharya, Blackwell and Sen 2016). Vertical bands are 95% confidence intervals.

Figure A10. Sensitivity of ACDE to Unobserved Mediator Confounding (Muzong Mediators)



Notes: The horizontal axis is ρ, the correlation between the mediator and the outcome error under an unobserved confounder. The vertical axis is the resulting ACDE. Shaded region is the 95% confidence band. Rows correspond to rebellion (top), mutiny (middle), and civilian uprising (bottom); columns correspond to the two mediators considered.

B Additional Tables

Table B1. Summary Statistics

Panel A. Summary Statistics of Province-Year Level Variables

Variable Name	**Mean**	**S.D.**	**Min**	**Max**	**N**
Rebellion (Binary)	0.008	0.086	0	1	4,785
Rebellion (Continuous)	0.008	0.086	0	1	4,785
Mutiny (Binary)	0.013	0.113	0	1	4,785
Mutiny (Continuous)	0.014	0.121	0	2	4,785
Uprising (Binary)	0.018	0.134	0	1	4,785
Uprising (Continuous)	0.033	0.306	0	7	4,785
Reform	0.568	0.495	0	1	4,785
Agricultural Disasters	0.08	0.35	0	6	4,785

Panel B. Summary Statistics of Province-Level Variables

Variable Name	**Mean**	**S.D.**	**Min**	**Max**	**N**
NumCCAPref	1.15	1.14	0	4.92	61
Longitude	112.33	4.22	103.51	120.59	61
Latitude	32.74	4.61	19.62	40.81	61
Terrain Ruggedness	132.92	113.13	2.22	481.77	61
Distance to Capital	638.22	358.62	82.92	1,627.11	61
River Density	0.19	0.06	0.05	0.33	61
Population	943,688.58	981,169.84	37,983.66	4,358,796	61

Panel C. Summary Statistics of Battle-Level Variables

Variable Name	**Mean**	**S.D.**	**Min**	**Max**	**N**
NumCCAPref	1.576	1.240	0	4.920	434
Central Army	0.039	0.194	0	1	434
Longitude of Battle Site	111.800	5.184	102.183	121.543	434
Latitude of Battle Site	32.714	4.277	20.024	40.387	434
Terrain Ruggedness of Battle Site	82.293	165.530	1.421	913.916	434
Battle Site's Distance to Capital City	660.516	343.321	5.650	1,578.625	434
Battle Site's Distance to Province	207.772	206.400	0.126	1,553.408	434
Rice Suitability of Battle Site	2,195.605	2,720.555	0	9,943.250	434
Millet Suitability of Battle Site	5,214.546	2,763.691	0	9,767.500	434
Wheat Suitability of Battle Site	4,181.936	1,795.240	0	8,806.750	434

Notes: This table presents the summary statistics of variables (before log-transformation) used in province-year level analysis (Panels A and B) and in battle-level analysis (Panel C).

Table B2. Accounting for Province Types

	Indicator for Rebellion			Indicator for Mutiny			Indicator for Uprising		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
NumCCAPref × Reform	-0.0047**	-0.0056**	-0.0091***	0.0114***	0.0128**	0.0170**	0.0216***	0.0157***	0.0144***
	(0.0023)	(0.0023)	(0.0025)	(0.0035)	(0.0055)	(0.0077)	(0.0037)	(0.0042)	(0.0044)
Agricultural Disasters		0.0005	0.0002		0.0007	0.0003		0.0031	0.0038
		(0.0012)	(0.0014)		(0.0026)	(0.0040)		(0.0025)	(0.0034)
Longitude × Reform		-0.0009	-0.0005		-8.88×10^{-5}	-0.0007		-0.0016	-0.0022
		(0.0009)	(0.0012)		(0.0016)	(0.0024)		(0.0012)	(0.0017)
Latitude × Reform		-0.0004	0.0021*		-0.0016	-0.0041		0.0012	0.0005
		(0.0006)	(0.0013)		(0.0010)	(0.0041)		(0.0008)	(0.0025)
Ruggedness × Reform		2.07×10^{-5}	-2.3×10^{-5}		4.38×10^{-5}	6.6×10^{-5}		-3.45×10^{-5}	-6.02×10^{-6}
		(2.74×10^{-5})	(3.19×10^{-5})		(3.78×10^{-5})	(4.12×10^{-5})		(3.02×10^{-5})	(4.14×10^{-5})
Log Distance to Capital × Reform		0.0062	-0.0034		0.0071	0.0198		0.0097*	0.0155
		(0.0050)	(0.0081)		(0.0081)	(0.0174)		(0.0053)	(0.0094)
River Density × Reform		-0.0224	0.0230		-0.0377	-0.0289		0.1459**	0.1345**
		(0.0448)	(0.0577)		(0.1071)	(0.1272)		(0.0613)	(0.0559)
Log Population × Reform		-0.0006	0.0001		-0.0011	0.0016		-0.0020	-0.0017
		(0.0023)	(0.0023)		(0.0052)	(0.0076)		(0.0033)	(0.0041)
Central Protection Provinces × Reform		0.0082	0.0102		0.0324**	0.0221		-0.0284***	-0.0042
		(0.0092)	(0.0077)		(0.0152)	(0.0327)		(0.0095)	(0.0131)
Northeastern Warlord Provinces × Reform		0.0106			0.0082			-0.0371**	
		(0.0108)			(0.0188)			(0.0171)	
Border Defense Provinces × Reform		-0.0137	-0.0174		0.0002	0.0192		-0.0387***	-0.0383*
		(0.0083)	(0.0107)		(0.0136)	(0.0283)		(0.0108)	(0.0196)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region-Year FE			Yes			Yes			Yes
Observations	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785
R-Squared	0.05783	0.05961	0.27793	0.04817	0.05146	0.22515	0.11870	0.12223	0.40958

Notes: Columns 2, 3, 5, 6, 8 and 9 control for the interaction between four province types and the reform dummy. The four types are: northeastern warlords, southeastern fiscal basket (baseline), central protection and border defense (Zhang 2010: 44-45). Note that the three autonomous provinces in the northeast are not in our dataset, and that the "northeastern warlords" category consists of other northeastern provinces that are not autonomous. Robust standard errors in parentheses are clustered at the province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B3. Accounting for the Effect of Fiscal Centralization in 809 CE

	Indicator for Rebellion			Indicator for Mutiny			Indicator for Uprising		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
NumCCAPref × Reform	-0.0047**	-0.0064**	-0.0078***	0.0114***	0.0112**	0.0173**	0.0216***	0.0204***	0.0155***
	(0.0023)	(0.0024)	(0.0026)	(0.0035)	(0.0051)	(0.0076)	(0.0037)	(0.0051)	(0.0046)
Agricultural Disasters		0.0007	0.0003		0.0011	0.0003		0.0034	0.0050
		(0.0012)	(0.0015)		(0.0027)	(0.0040)		(0.0026)	(0.0034)
Longitude × Reform		0.0004	-0.0003		0.0006	-0.0009		-0.0001	-0.0021*
		(0.0010)	(0.0013)		(0.0014)	(0.0023)		(0.0011)	(0.0012)
Latitude × Reform		9.8×10^{-5}	0.0010		-0.0011	-0.0040		-0.0001	-0.0007
		(0.0006)	(0.0011)		(0.0011)	(0.0036)		(0.0009)	(0.0022)
Ruggedness × Reform		2.11×10^{-5}	-2.08×10^{-5}		4.6×10^{-6}	5.26×10^{-5}		2.26×10^{-5}	1.11×10^{-5}
		(2.32×10^{-5})	(3.28×10^{-5})		(3.47×10^{-5})	(3.81×10^{-5})		(2.66×10^{-5})	(4.47×10^{-5})
Log Distance to Capital × Reform		0.0056	6.42×10^{-5}		0.0003	0.0135		0.0136*	0.0265***
		(0.0054)	(0.0080)		(0.0087)	(0.0148)		(0.0071)	(0.0092)
River Density × Reform		0.0086	0.0236		-0.0047	-0.0203		0.1550**	0.1261**
		(0.0480)	(0.0638)		(0.1235)	(0.1335)		(0.0600)	(0.0568)
Log Population × Reform		0.0011	0.0016		-0.0005	-0.0032		0.0016	0.0058
		(0.0031)	(0.0032)		(0.0067)	(0.0106)		(0.0042)	(0.0049)
Fiscal Centralization × Reform		0.0001	-0.0009		0.0008	0.0016		-0.0023*	-0.0031**
		(0.0010)	(0.0009)		(0.0015)	(0.0023)		(0.0013)	(0.0013)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region-Year FE			Yes			Yes			Yes
Observations	4,785	4,516	4,516	4,785	4,516	4,516	4,785	4,516	4,516
R-Squared	0.05783	0.05418	0.27588	0.04817	0.04900	0.24208	0.11870	0.12555	0.41396

Notes: Columns 2, 3, 5, 6, 8 and 9 control for the interaction between the number of prefectures in each province in 809 CE (denoted as "Fiscal Centralization") and the reform dummy. Robust standard errors in parentheses are clustered at the province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B4. Accounting for the "Two-Tax Policy"

	Indicator for Rebellion			Indicator for Mutiny			Indicator for Uprising		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
NumCCAPref × Reform	-0.0047**	-0.0059**	-0.0077***	0.0114***	0.0125**	0.0164**	0.0216***	0.0171***	0.0156***
	(0.0023)	(0.0025)	(0.0024)	(0.0035)	(0.0053)	(0.0075)	(0.0037)	(0.0045)	(0.0041)
Agricultural Disasters		0.0006	0.0002		0.0008	0.0002		0.0030	0.0038
		(0.0012)	(0.0014)		(0.0026)	(0.0040)		(0.0025)	(0.0034)
Longitude × Reform		0.0007	-0.0002		0.0011	-0.0019		-0.0011	-0.0020
		(0.0009)	(0.0012)		(0.0014)	(0.0020)		(0.0009)	(0.0015)
Latitude × Reform		-3.21×10^{-5}	0.0009		-0.0016	-0.0032		0.0012	-0.0009
		(0.0006)	(0.0012)		(0.0010)	(0.0034)		(0.0008)	(0.0017)
Ruggedness × Reform		2.77×10^{-5}	-2.53×10^{-5}		2.63×10^{-5}	6×10^{-5}		-3.33×10^{-5}	-8.6×10^{-6}
		(2.68×10^{-5})	(3.08×10^{-5})		(3.91×10^{-5})	(4.24×10^{-5})		(2.9×10^{-5})	(4.24×10^{-5})
Log Distance to Capital × Reform		0.0057	-0.0015		0.0024	0.0158		0.0079	0.0199**
		(0.0045)	(0.0078)		(0.0081)	(0.0156)		(0.0051)	(0.0089)
River Density × Reform		0.0057	0.0300		-0.0093	-0.0306		0.1685***	0.1499**
		(0.0476)	(0.0629)		(0.1157)	(0.1355)		(0.0532)	(0.0584)
Log Population × Reform		0.0013	-0.0006		0.0010	0.0008		-0.0026	-0.0024
		(0.0022)	(0.0025)		(0.0051)	(0.0074)		(0.0031)	(0.0036)
Two Tax × Reform		-0.0050	0.0085		-0.0139	0.0031		0.0352***	0.0372**
		(0.0078)	(0.0085)		(0.0122)	(0.0164)		(0.0083)	(0.0149)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region-Year FE			Yes			Yes			Yes
Observations	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785
R-Squared	0.05783	0.05865	0.27768	0.04817	0.04973	0.22482	0.11870	0.12226	0.40980

Notes: Columns 2, 3, 5, 6, 8 and 9 control for the interaction between a dummy variable for the provinces where the Two-Tax Policy was implemented according to Peterson (1979) (p. 520) and the reform dummy. Robust standard errors in parentheses are clustered at the province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B5. Alternative Measure of Reform Exposure

	Indicator for Rebellion			Indicator for Mutiny			Indicator for Uprising		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
CCAPref (Binary) × Reform	-0.0128**	-0.0140**	-0.0106*	0.0186**	0.0193**	0.0240*	0.0355***	0.0327***	0.0215**
	(0.0049)	(0.0059)	(0.0055)	(0.0088)	(0.0084)	(0.0122)	(0.0081)	(0.0078)	(0.0086)
Agricultural Disasters		0.0006	0.0002		0.0009	0.0003		0.0031	0.0040
		(0.0012)	(0.0014)		(0.0026)	(0.0040)		(0.0025)	(0.0034)
Longitude × Reform		0.0005	-6.28×10^{-5}		-0.0002	-0.0013		0.0006	2.5×10^{-5}
		(0.0007)	(0.0010)		(0.0009)	(0.0021)		(0.0010)	(0.0020)
Latitude × Reform		-0.0003	5.72×10^{-5}		-0.0008	-0.0021		0.0015	-0.0006
		(0.0006)	(0.0012)		(0.0010)	(0.0030)		(0.0011)	(0.0020)
Ruggedness × Reform		7.13×10^{-6}	-2.46×10^{-5}		2.97×10^{-5}	6.19×10^{-5}		4.87×10^{-5}	-6.48×10^{-6}
		(2.57×10^{-5})	(2.93×10^{-5})		(3.64×10^{-5})	(3.81×10^{-5})		(3.51×10^{-5})	(6.2×10^{-5})
Log Distance to Capital × Reform		0.0014	-0.0032		0.0093	0.0205		0.0188***	0.0249**
		(0.0052)	(0.0077)		(0.0085)	(0.0180)		(0.0061)	(0.0108)
River Density × Reform		-0.0072	0.0146		0.0098	0.0003		0.1956***	0.1680**
		(0.0460)	(0.0663)		(0.1251)	(0.1507)		(0.0619)	(0.0772)
Log Population × Reform		0.0011	-0.0015		0.0030	0.0031		0.0003	0.0010
		(0.0026)	(0.0025)		(0.0042)	(0.0062)		(0.0029)	(0.0039)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region-Year FE			Yes			Yes			Yes
Observations	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785
R-Squared	0.05828	0.05876	0.27732	0.04730	0.04909	0.22399	0.11653	0.11992	0.40830

Notes: CCAPref (Binary) is a binary variable that is equal to 1(0) when NumCCAPref is above(below) the median number of CCA prefectures. Robust standard errors in parentheses are clustered at the province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B6. Number of Rebellions, Mutinies, and Uprisings

	Number of Rebellions			Number of Mutinies			Number of Uprisings		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
NumCCAPref × Reform	-0.0047**	-0.0063**	-0.0073***	0.0115***	0.0111**	0.0167**	0.0411***	0.0361**	0.0267***
	(0.0023)	(0.0025)	(0.0024)	(0.0037)	(0.0052)	(0.0074)	(0.0107)	(0.0138)	(0.0087)
Agricultural Disasters		0.0006	0.0002		0.0005	-0.0002		0.0052	0.0098
		(0.0012)	(0.0014)		(0.0026)	(0.0040)		(0.0077)	(0.0101)
Longitude × Reform		0.0003	0.0001		0.0001	-0.0019		0.0042	0.0087**
		(0.0007)	(0.0011)		(0.0010)	(0.0019)		(0.0027)	(0.0041)
Latitude × Reform		5.6×10^{-5}	0.0006		-0.0013	-0.0037		0.0004	-0.0098**
		(0.0006)	(0.0011)		(0.0010)	(0.0034)		(0.0016)	(0.0038)
Ruggedness × Reform		2.12×10^{-5}	-2.4×10^{-5}		2.51×10^{-6}	6.22×10^{-5}		-3.11×10^{-5}	-3.97×10^{-5}
		(2.27×10^{-5})	(3.01×10^{-5})		(3.51×10^{-5})	(4.22×10^{-5})		(5.6×10^{-5})	(7.34×10^{-5})
Log Distance to Capital × Reform		0.0058	-0.0012		0.0029	0.0187		0.0082	0.0319*
		(0.0046)	(0.0076)		(0.0083)	(0.0164)		(0.0112)	(0.0182)
River Density × Reform		0.0061	0.0285		0.0170	-0.0040		0.2355*	0.1065
		(0.0474)	(0.0645)		(0.1232)	(0.1418)		(0.1178)	(0.1200)
Log Population × Reform		0.0013	-0.0005		0.0013	0.0011		-0.0047	-0.0084
		(0.0023)	(0.0026)		(0.0051)	(0.0071)		(0.0082)	(0.0103)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region-Year FE			Yes			Yes			Yes
Observations	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785
R-Squared	0.05783	0.05858	0.27760	0.04985	0.05087	0.22277	0.09480	0.09673	0.38748

Notes: This table displays results using the number of rebellions, mutinies and uprisings as outcome variables. Robust standard errors in parentheses are clustered at the province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B7. Subsample Excluding Zelu Province

	Indicator for Rebellion			Indicator for Mutiny			Indicator for Uprising		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
NumCCAPref × Reform	-0.0050**	-0.0061**	-0.0076***	0.0107***	0.0118**	0.0132*	0.0221***	0.0198***	0.0181***
	(0.0023)	(0.0025)	(0.0025)	(0.0033)	(0.0051)	(0.0068)	(0.0038)	(0.0048)	(0.0042)
Agricultural Disasters		0.0006	0.0002		0.0009	-0.0003		0.0031	0.0040
		(0.0012)	(0.0014)		(0.0027)	(0.0041)		(0.0026)	(0.0035)
Longitude × Reform		0.0003	0.0002		4.26×10^{-5}	-0.0014		0.0013	-0.0006
		(0.0007)	(0.0011)		(0.0010)	(0.0019)		(0.0009)	(0.0020)
Latitude × Reform		-6.13×10^{-5}	0.0007		-0.0017*	-0.0020		0.0007	-0.0024
		(0.0006)	(0.0012)		(0.0009)	(0.0029)		(0.0010)	(0.0019)
Ruggedness × Reform		2.17×10^{-5}	-2.35×10^{-5}		8.86×10^{-6}	6.76×10^{-5}*		1.24×10^{-5}	-4.75×10^{-6}
		(2.22×10^{-5})	(3.03×10^{-5})		(3.61×10^{-5})	(3.76×10^{-5})		(2.83×10^{-5})	(3.76×10^{-5})
Log Distance to Capital × Reform		0.0059	-0.0009		0.0027	0.0202		0.0072	0.0204**
		(0.0046)	(0.0078)		(0.0079)	(0.0149)		(0.0061)	(0.0085)
River Density × Reform		0.0216	0.0348		0.0353	0.0595		0.1433**	0.1165*
		(0.0467)	(0.0712)		(0.1147)	(0.1324)		(0.0553)	(0.0617)
Log Population × Reform		0.0007	-0.0005		-0.0006	0.0007		-0.0022	-0.0019
		(0.0022)	(0.0027)		(0.0049)	(0.0071)		(0.0033)	(0.0036)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region-Year FE			Yes			Yes			Yes
Observations	4,676	4,676	4,676	4,676	4,676	4,676	4,676	4,676	4,676
R-Squared	0.06177	0.06259	0.29617	0.04718	0.04888	0.22864	0.12191	0.12401	0.40998

Notes: This table displays results excluding the Zelu province. Robust standard errors in parentheses are clustered at the province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B8. Number of Rebellions, Mutinies, and Uprisings, Weighted by Provincial Population

	Number of Rebellions, Population-Weighted			Number of Mutinies, Population-Weighted			Number of Uprisings, Population-Weighted		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
NumCCAPref × Reform	-0.0003*	-0.0005**	-0.0005***	0.0008***	0.0008**	0.0013**	0.0031***	0.0027**	0.0020***
	(0.0002)	(0.0002)	(0.0002)	(0.0003)	(0.0004)	(0.0005)	(0.0008)	(0.0010)	(0.0007)
Agricultural Disasters		5.1×10^{-5}	2.36×10^{-5}		4.16×10^{-5}	-1.4×10^{-5}		0.0005	0.0008
		(9.19×10^{-5})	(0.0001)		(0.0002)	(0.0003)		(0.0006)	(0.0008)
Longitude × Reform		2.53×10^{-5}	2.22×10^{-5}		9.32×10^{-7}	-0.0002		0.0003	0.0007**
		(5.24×10^{-5})	(8.08×10^{-5})		(7.99×10^{-5})	(0.0001)		(0.0002)	(0.0003)
Latitude × Reform		1.22×10^{-7}	5.5×10^{-5}		-0.0001	-0.0003		5.31×10^{-5}	-0.0007**
		(4.6×10^{-5})	(8.72×10^{-5})		(7.96×10^{-5})	(0.0003)		(0.0001)	(0.0003)
Ruggedness × Reform		1.96×10^{-6}	-1.37×10^{-6}		-1.34×10^{-7}	4.52×10^{-6}		-1.87×10^{-6}	-2.08×10^{-6}
		(1.75×10^{-6})	(2.28×10^{-6})		(2.64×10^{-6})	(3.14×10^{-6})		(4.31×10^{-6})	(5.54×10^{-6})
Log Distance to Capital × Reform		0.0005	-0.0001		0.0002	0.0014		0.0005	0.0024*
		(0.0003)	(0.0006)		(0.0006)	(0.0012)		(0.0008)	(0.0014)
River Density × Reform		0.0007	0.0019		0.0026	0.0007		0.0154*	0.0056
		(0.0036)	(0.0049)		(0.0090)	(0.0102)		(0.0089)	(0.0097)
Log Population × Reform		0.0001	3.11×10^{-6}		0.0001	9.73×10^{-5}		-0.0003	-0.0006
		(0.0002)	(0.0002)		(0.0004)	(0.0005)		(0.0006)	(0.0008)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region-Year FE			Yes			Yes			Yes
Observations	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785
R-Squared	0.05629	0.05733	0.26948	0.04762	0.04892	0.21758	0.09153	0.09354	0.37434

Notes: This table displays results using as outcome variables the number of rebellions, mutinies and uprisings reweighted by provincial population. Robust standard errors in parentheses are clustered at the province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B9. Accounting for Border Regions

	Indicator for Rebellion			Indicator for Mutiny			Indicator for Uprising		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
NumCCAPref × Reform	-0.0047**	-0.0074***	-0.0080***	0.0114***	0.0121**	0.0169**	0.0216***	0.0190***	0.0164***
	(0.0023)	(0.0023)	(0.0023)	(0.0035)	(0.0060)	(0.0078)	(0.0037)	(0.0048)	(0.0040)
Agricultural Disasters		0.0006	0.0002		0.0009	0.0002		0.0030	0.0039
		(0.0012)	(0.0014)		(0.0026)	(0.0040)		(0.0025)	(0.0034)
Longitude × Reform		-0.0001	-0.0007		0.0004	-0.0013		0.0009	-0.0015
		(0.0008)	(0.0013)		(0.0015)	(0.0032)		(0.0010)	(0.0019)
Latitude × Reform		0.0006	0.0013		-0.0017	-0.0037		0.0010	-0.0013
		(0.0008)	(0.0013)		(0.0014)	(0.0036)		(0.0008)	(0.0020)
Ruggedness × Reform		2.29×10^{-5}	-1.97×10^{-5}		7.24×10^{-6}	5.81×10^{-5}		1.35×10^{-5}	1.62×10^{-6}
		(2.28×10^{-5})	(2.98×10^{-5})		(3.68×10^{-5})	(3.97×10^{-5})		(2.81×10^{-5})	(3.94×10^{-5})
Log Distance to Capital × Reform		0.0078	0.0008		0.0013	0.0148		0.0091	0.0236**
		(0.0047)	(0.0074)		(0.0100)	(0.0166)		(0.0056)	(0.0089)
River Density × Reform		-0.0021	0.0106		-0.0024	-0.0212		0.1580***	0.1246**
		(0.0522)	(0.0766)		(0.1143)	(0.1316)		(0.0573)	(0.0563)
Log Population × Reform		0.0020	0.0006		0.0008	0.0002		-0.0026	-0.0007
		(0.0024)	(0.0030)		(0.0052)	(0.0088)		(0.0034)	(0.0038)
Border × Reform		-0.0067	-0.0081		0.0046	0.0045		-0.0061	-0.0085
		(0.0067)	(0.0084)		(0.0123)	(0.0172)		(0.0069)	(0.0060)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region-Year FE			Yes			Yes			Yes
Observations	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785
R-Squared	0.05783	0.05874	0.27776	0.04817	0.04947	0.22484	0.11870	0.12092	0.40923

Notes: Columns 2, 3, 5, 6, 8 and 9 control for the interaction between border regions and the reform dummy. Robust standard errors in parentheses are clustered at the province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B10. Accounting for Pre-Reform Tang Victories over Provinces

	Indicator for Rebellion			Indicator for Mutiny			Indicator for Uprising		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
NumCCAPref × Reform	-0.0047**	-0.0067***	-0.0054*	0.0114***	0.0116*	0.0182**	0.0216***	0.0204***	0.0178***
	(0.0023)	(0.0022)	(0.0029)	(0.0035)	(0.0060)	(0.0089)	(0.0037)	(0.0048)	(0.0048)
Agricultural Disasters		0.0006	0.0009		0.0008	-5.86×10^{-6}		0.0026	0.0029
		(0.0012)	(0.0014)		(0.0028)	(0.0046)		(0.0026)	(0.0037)
Longitude × Reform		-0.0006	-0.0006		-0.0012	-0.0026		0.0014	-0.0009
		(0.0006)	(0.0009)		(0.0011)	(0.0027)		(0.0011)	(0.0024)
Latitude × Reform		0.0007	0.0004		-0.0012	-0.0039		0.0009	-0.0015
		(0.0006)	(0.0016)		(0.0011)	(0.0045)		(0.0011)	(0.0022)
Ruggedness × Reform		5.43×10^{-6}	2.82×10^{-5}		-3.38×10^{-5}	6.83×10^{-5}		3.2×10^{-5}	3.47×10^{-5}
		(2.3×10^{-5})	(4.79×10^{-5})		(5.67×10^{-5})	(9.99×10^{-5})		(4.36×10^{-5})	(7.79×10^{-5})
Log Distance to Capital × Reform		0.0078*	0.0059		0.0060	0.0223		0.0058	0.0222**
		(0.0044)	(0.0090)		(0.0094)	(0.0205)		(0.0067)	(0.0106)
River Density × Reform		-0.0377	0.0134		-0.0388	-0.0783		0.1660***	0.1419**
		(0.0503)	(0.0764)		(0.1305)	(0.1595)		(0.0594)	(0.0645)
Log Population × Reform		0.0037	0.0013		0.0046	0.0021		-0.0036	-0.0006
		(0.0025)	(0.0032)		(0.0066)	(0.0095)		(0.0045)	(0.0054)
Pre-Reform Victory × Reform		-0.0212***	-0.0183**		-0.0118	-0.0140		-0.0022	-0.0114
		(0.0057)	(0.0068)		(0.0141)	(0.0207)		(0.0090)	(0.0071)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region-Year FE			Yes			Yes			Yes
Observations	4,785	4,071	4,071	4,785	4,071	4,071	4,785	4,071	4,071
R-Squared	0.05783	0.06321	0.33582	0.04817	0.05308	0.24647	0.11870	0.12366	0.43306

Notes: Columns 2, 3, 5, 6, 8 and 9 control for the interaction between pre-Reform Tang victories over provinces and the reform dummy. Robust standard errors in parentheses are clustered at the province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B11. Alternative Coding of Reform Exposure

	Indicator for Rebellion			Indicator for Mutiny			Indicator for Uprising		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
NumCCAPref × Reform	-0.0047**	-0.0065***	-0.0069***	0.0099***	0.0088*	0.0155**	0.0188***	0.0162***	0.0140***
	(0.0019)	(0.0019)	(0.0020)	(0.0035)	(0.0050)	(0.0070)	(0.0038)	(0.0047)	(0.0041)
Agricultural Disasters		0.0005	0.0002		0.0009	0.0002		0.0031	0.0039
		(0.0012)	(0.0014)		(0.0026)	(0.0040)		(0.0025)	(0.0034)
Longitude × Reform		0.0002	9.85×10^{-5}		0.0002	-0.0017		0.0014	-0.0003
		(0.0007)	(0.0010)		(0.0011)	(0.0019)		(0.0010)	(0.0020)
Latitude × Reform		4.26×10^{-5}	0.0005		-0.0012	-0.0031		0.0007	-0.0015
		(0.0006)	(0.0011)		(0.0010)	(0.0032)		(0.0010)	(0.0020)
Ruggedness × Reform		2.16×10^{-5}	-2.55×10^{-5}		9.74×10^{-6}	6.35×10^{-5}		1.4×10^{-5}	-4.67×10^{-6}
		(2.19×10^{-5})	(2.66×10^{-5})		(3.7×10^{-5})	(4.04×10^{-5})		(2.89×10^{-5})	(4.86×10^{-5})
Log Distance to Capital × Reform		0.0066	-0.0007		0.0023	0.0148		0.0065	0.0198*
		(0.0044)	(0.0070)		(0.0085)	(0.0151)		(0.0069)	(0.0102)
River Density × Reform		0.0135	0.0429		-0.0186	-0.0636		0.1461**	0.1108
		(0.0463)	(0.0652)		(0.1221)	(0.1342)		(0.0613)	(0.0684)
Log Population × Reform		0.0016	-0.0005		0.0023	0.0009		-0.0016	-0.0010
		(0.0023)	(0.0026)		(0.0050)	(0.0071)		(0.0034)	(0.0037)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region-Year FE			Yes			Yes			Yes
Observations	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785
R-Squared	0.05792	0.05875	0.27762	0.04788	0.04904	0.22482	0.11800	0.12017	0.40880

Notes: Number of CCA prefectures is computed using official histories only. Robust standard errors in parentheses are clustered at the province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B12. Effects of Reform on the Number of Battles

	Number of Battles		
	(1)	(2)	(3)
NumCCAPref × Reform	-0.0172	-0.0072	-0.0173
	(0.0184)	(0.0129)	(0.0154)
Agricultural Disasters		0.0025	0.0020
		(0.0071)	(0.0087)
Longitude × Reform		0.0143**	0.0166**
		(0.0057)	(0.0079)
Latitude × Reform		-0.0042	0.0093
		(0.0039)	(0.0085)
Ruggedness × Reform		0.0002	2.96×10^{-5}
		(0.0002)	(0.0002)
Log Distance to Capital × Reform		-0.0153	-0.0940*
		(0.0247)	(0.0520)
River Density × Reform		-0.5739*	-1.049**
		(0.3048)	(0.4407)
Log Population × Reform		-0.0180	-0.0082
		(0.0168)	(0.0196)
Year FE	Yes	Yes	Yes
Province FE	Yes	Yes	Yes
Region-Year FE			Yes
Observations	4,785	4,785	4,785
R-Squared	0.10922	0.11269	0.41126

Notes: This table displays the effects of military centralization on the number of battles in which provinces participate. Robust standard errors in parentheses are clustered at the province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B13. Effects of Reform on the Probability of Uprising, Headquarters vs. Outer Prefectures

	Indicator for Uprising			Indicator for Headquarter Uprising			Indicator for Outer Prefecture Uprising		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
NumCCAPref × Reform	0.0216***	0.0200***	0.0171***	0.0098***	0.0091***	0.0091***	0.0146***	0.0133***	0.0101***
	(0.0037)	(0.0047)	(0.0041)	(0.0026)	(0.0023)	(0.0030)	(0.0037)	(0.0047)	(0.0034)
Agricultural Disasters		0.0030	0.0039		0.0025	0.0032		-0.0002	0.0002
		(0.0025)	(0.0034)		(0.0020)	(0.0025)		(0.0017)	(0.0026)
Longitude × Reform		0.0013	-0.0005		0.0012*	-0.0013		0.0005	0.0008
		(0.0009)	(0.0020)		(0.0007)	(0.0014)		(0.0009)	(0.0015)
Latitude × Reform		0.0006	-0.0020		0.0003	-0.0018		-0.0002	-0.0031*
		(0.0010)	(0.0019)		(0.0008)	(0.0013)		(0.0006)	(0.0016)
Ruggedness × Reform		1.2×10^{-5}	-2.99×10^{-6}		1×10^{-5}	-1.91×10^{-5}		-7.39×10^{-6}	-5.43×10^{-6}
		(2.81×10^{-5})	(3.85×10^{-5})		(2.3×10^{-5})	(3.38×10^{-5})		(2.3×10^{-5})	(2.7×10^{-5})
Log Distance to Capital × Reform		0.0073	0.0216**		0.0030	0.0117		0.0043	0.0165**
		(0.0058)	(0.0087)		(0.0043)	(0.0078)		(0.0055)	(0.0074)
River Density × Reform		0.1654***	0.1435**		0.0934**	0.0793		0.1439***	0.1091*
		(0.0570)	(0.0616)		(0.0382)	(0.0600)		(0.0539)	(0.0566)
Log Population × Reform		-0.0031	-0.0019		-0.0025	-0.0036		-0.0013	0.0002
		(0.0033)	(0.0036)		(0.0022)	(0.0029)		(0.0028)	(0.0024)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region-Year FE			Yes			Yes			Yes
Observations	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785
R-Squared	0.11870	0.12086	0.40916	0.08677	0.08886	0.29871	0.09250	0.09383	0.40408

Notes: Columns 4 to 9 decompose all uprisings into uprisings that occurred in the headquarters and in outer prefectures separately. Robust standard errors in parentheses are clustered at the province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B14. Effect of Reform on Economic Development

	Log Number of Households		
	(1)	(2)	(3)
NumCCAPref × Reform	0.1599	0.2917	0.3874
	(0.2526)	(0.3658)	(0.3053)
Ruggedness × Reform		0.0046	0.0033
		(0.0032)	(0.0035)
Log Distance to Capital × Reform		-0.1046	0.2543
		(0.3418)	(0.5565)
River Density × Reform		11.16	14.52
		(11.55)	(9.316)
Rice × Reform		-0.0001	-0.0010*
		(0.0003)	(0.0005)
Wheat × Reform		-1.54×10^{-5}	-7.89×10^{-5}
		(0.0004)	(0.0004)
Millet × Reform		4.02×10^{-5}	0.0004
		(0.0001)	(0.0003)
Post Dummy	Yes	Yes	
Province FE	Yes	Yes	Yes
Region-Post Dummy			Yes
Observations	76	76	76
R-Squared	0.80422	0.85055	0.92043
Within R-Squared	0.01382	0.24719	0.43318

Notes: This table uses the log number of households in mid-Tang and early Northern Song (approximately 741 CE and 980 CE respectively) as a proxy for economic development. Robust standard errors in parentheses are clustered at the province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B15. Civilian Uprisings: Initial vs. Subsequent

	Indicator for Newly Initiated Uprising			Indicator for Uprising Spreading In		
	(1)	(2)	(3)	(4)	(5)	(6)
NumCCAPref × Reform	0.0098***	0.0093**	0.0080**	0.0144***	0.0125***	0.0092***
	(0.0033)	(0.0036)	(0.0033)	(0.0022)	(0.0025)	(0.0018)
Agricultural Disasters		0.0058**	0.0064*		-0.0011	-0.0006
		(0.0024)	(0.0033)		(0.0015)	(0.0020)
Longitude × Reform		-0.0013	-0.0014		0.0025***	0.0025***
		(0.0008)	(0.0020)		(0.0005)	(0.0007)
Latitude × Reform		0.0008	-0.0019		-0.0001	-0.0007
		(0.0006)	(0.0014)		(0.0006)	(0.0010)
Ruggedness × Reform		-1.63×10^{-6}	-1.04×10^{-5}		1.24×10^{-5}	-1.09×10^{-5}
		(2.02×10^{-5})	(3.71×10^{-5})		(1.83×10^{-5})	(2.11×10^{-5})
Log Distance to Capital × Reform		0.0079**	0.0157*		0.0005	0.0025
		(0.0036)	(0.0084)		(0.0047)	(0.0040)
River Density × Reform		0.0615	0.0575		0.0880*	0.0404
		(0.0385)	(0.0571)		(0.0452)	(0.0478)
Log Population × Reform		-5.94×10^{-5}	-0.0035		-0.0019	0.0005
		(0.0025)	(0.0028)		(0.0020)	(0.0019)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Region-Year FE			Yes			Yes
Observations	4,785	4,785	4,785	4,785	4,785	4,785
R-Squared	0.05262	0.05722	0.28640	0.13391	0.13660	0.44263

Notes: Columns 1 to 3 display results for civilian uprisings that were newly initiated, and columns 4 to 6 display results for civilian uprisings that subsequently spread in from other places. Robust standard errors in parentheses are clustered at the province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B16. Correlates of CCA

	Number of CCA Prefectures								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
#Battles during An Lushan Rebellion (standardized)	0.631***								0.501***
	(0.124)								(0.146)
Longitude		0.060*							0.010
		(0.035)							(0.034)
Latitude			0.018						0.001
			(0.024)						(0.024)
Ruggedness				-0.001					0.000
				(0.001)					(0.001)
Log Distance to Capital					0.094				-0.024
					(0.162)				(0.211)
River Density						3.760**			1.669
						(1.742)			(1.977)
Log Population							0.463***		0.207*
							(0.088)		(0.107)
#Prefectures in Province								0.058	0.054
								(0.037)	(0.042)
Observations	61	61	61	61	61	61	61	61	61
R-Squared	0.399	0.063	0.007	0.022	0.004	0.048	0.316	0.052	0.531

Notes: This table shows the correlation between the number of CCA prefectures with the control variables used. Robust standard errors in parentheses. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B17. Effect of Reform on Commissioner Appointment and Characteristics

	Reappointed Elsewhere	Non-Chinese	*Keju* Degree	Military Merit	Father Official	Father Official, Not Local	Father Official, Not Centrally-Appointed Local
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
NumCCAPref × Reform	0.029*	0.018	-0.004	0.004	0.047	-0.020	0.029
	(0.015)	(0.033)	(0.035)	(0.036)	(0.034)	(0.053)	(0.061)
Agricultural Disasters	-0.013	0.011	0.005	0.003	0.0010	-0.002	-0.003
	(0.010)	(0.006)	(0.009)	(0.007)	(0.011)	(0.010)	(0.010)
Longitude × Reform	-0.013	-0.005	-0.027	0.024	-0.023	-0.044**	-0.050**
	(0.008)	(0.013)	(0.019)	(0.018)	(0.017)	(0.022)	(0.023)
Latitude × Reform	0.008	-0.009	0.057***	0.011	0.017	0.062**	0.030
	(0.009)	(0.018)	(0.018)	(0.017)	(0.015)	(0.028)	(0.028)
Ruggedness × Reform	0.0002	0.0004	0.0009**	-0.0005	0.001**	0.001	0.001
	(0.0002)	(0.0006)	(0.0004)	(0.0005)	(0.0006)	(0.001)	(0.001)
Log Distance to Capital × Reform	0.048	0.149	0.041	-0.106	0.123	0.270*	0.162
	(0.049)	(0.114)	(0.096)	(0.125)	(0.100)	(0.150)	(0.159)
River Density × Reform	0.581**	1.58	1.30	-1.58	0.749	1.51	0.905
	(0.271)	(1.04)	(0.792)	(1.11)	(0.626)	(1.12)	(1.32)
Log Population × Reform	0.048**	0.046	0.244***	-0.059	0.061	0.232***	0.166**
	(0.020)	(0.057)	(0.060)	(0.047)	(0.048)	(0.071)	(0.074)
Mean of Outcome	0.241	0.124	0.466	0.235	0.955	0.843	0.891
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,018	3,961	3,961	3,961	3,961	3,961	3,961
R-squared	0.358	0.418	0.528	0.590	0.399	0.468	0.419

Notes: This table shows the effect of Reform on commissioner-related outcomes. For a given year and province, the D.V. in column 1 is the number of commissioners who were reappointed to another position by the central government; the D.V. in column 2 is the number of non-Chinese commissioners; the D.V. in column 3 is the number of commissioners who had civil examination (*keju*) degrees; the D.V. in column 4 is the number of commissioners who entered officialdom through military merit; the D.V. in column 5 is the number of commissioners whose father was a government official; the D.V. in column 6 is the number of commissioners whose father was a government official, and who are not from the local province; the D.V. in column 7 is the number of commissioners whose father was a government official, and who are not centrally-appointed locals. Robust standard errors in parentheses are clustered at province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B18. Non-Results for Commissioner Characteristics

	Indicator for Rebellion (1)	Indicator for Mutiny (2)	Indicator for Uprising (3)
Reappointed Elsewhere	0.005	-0.0005	0.004
	(0.008)	(0.006)	(0.004)
Province Size	0.0009	0.003*	0.004*
	(0.002)	(0.001)	(0.002)
Agricultural Disasters	0.0003	-4.69×10^{-5}	0.002
	(0.002)	(0.005)	(0.003)
NumCCAPref × Reform	-0.007***	0.016*	0.016***
	(0.003)	(0.009)	(0.005)
Longitude × Reform	6.76×10^{-5}	-0.003	-0.0007
	(0.001)	(0.003)	(0.003)
Latitude × Reform	0.0004	-0.003	-0.002
	(0.001)	(0.005)	(0.003)
Ruggedness × Reform	-3.55×10^{-5}	4.2×10^{-5}	7.74×10^{-6}
	(4.01×10^{-5})	(6.67×10^{-5})	(6.23×10^{-5})
Log Distance to Capital × Reform	-0.001	0.020	0.021*
	(0.009)	(0.019)	(0.013)
River Density × Reform	0.015	-0.072	0.097
	(0.075)	(0.156)	(0.081)
Log Population × Reform	-0.002	0.002	0.003
	(0.003)	(0.012)	(0.007)
Year FE	Yes	Yes	Yes
Province FE	Yes	Yes	Yes
Region-Year FE	Yes	Yes	Yes
Observations	4,018	4,018	4,018
R-squared	0.299	0.254	0.471

Notes: This table shows the effects on the probabilities of rebellion, mutiny, and civilian uprisings of a commissioner-related indicator that was likely affected by the Reform according to Table B17: the number of commissioners who were centrally-reappointed to other positions. The specification includes the Reform dummy interacted with the number of CCA prefectures and all the control variables used in Section 4.1. The results, available upon request, are similar without the Reform treatment in the regressions or without other controls. Robust standard errors in parentheses are clustered at province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B19. Results for Successful Uprisings

	Pr(Successful Uprising)		
	(1)	(2)	(3)
NumCCAPref × Reform	0.0144***	0.0125***	0.0092***
	(0.0022)	(0.0025)	(0.0018)
Agricultural Disasters		-0.0011	-0.0006
		(0.0015)	(0.0020)
Longitude × Reform		0.0025***	0.0025***
		(0.0005)	(0.0007)
Latitude × Reform		-0.0001	-0.0007
		(0.0006)	(0.0010)
Ruggedness × Reform		1.24×10^{-5}	-1.09×10^{-5}
		(1.83×10^{-5})	(2.11×10^{-5})
Log Distance to Capital × Reform		0.0005	0.0025
		(0.0047)	(0.0040)
River Density × Reform		0.0880*	0.0404
		(0.0452)	(0.0478)
Log Population × Reform		-0.0019	0.0005
		(0.0020)	(0.0019)
Year FE	Yes	Yes	Yes
Province FE	Yes	Yes	Yes
Region-Year FE			Yes
Observations	4,785	4,785	4,785
R-Squared	0.13391	0.13660	0.44263

Notes: Robust standard errors in parentheses are clustered at province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B20. Results for Domestic Battles

	Indicator of Battle Victory for Tang		
	(1)	(2)	(3)
NumCCAPref × Reform	-0.138***	-0.167***	-0.162***
	(0.046)	(0.044)	(0.044)
Reform × Foreign War	0.028	-0.546***	-0.647***
	(0.134)	(0.196)	(0.223)
NumCCAPref × Reform × Foreign War	0.129	0.145	0.154
	(0.080)	(0.095)	(0.100)
Central Army		-0.206*	-0.208*
		(0.116)	(0.115)
Longitude of Battle Site		0.021	0.023
		(0.018)	(0.020)
Latitude of Battle Site		-0.053**	-0.058***
		(0.020)	(0.021)
Ruggedness of Battle Site		-0.0003*	-0.0004*
		(0.0002)	(0.0002)
Log Battle Site's Distance to Capital City		-0.034	-0.036
		(0.062)	(0.063)
Log Battle Site's Distance to Province		-0.014	-0.014
		(0.017)	(0.018)
Rice Suitability of Battle Site		6.34×10^{-6}	6.51×10^{-6}
		(1.52×10^{-5})	(1.54×10^{-5})
Millet Suitability of Battle Site		$4.35 \times 10^{-5*}$	$4.58 \times 10^{-5**}$
		(2.22×10^{-5})	(2.16×10^{-5})
Wheat Suitability of Battle Site		$-9.05 \times 10^{-5***}$	$-9.5 \times 10^{-5***}$
		(3.13×10^{-5})	(3.22×10^{-5})
Longitude × Reform		-0.065***	-0.070***
		(0.016)	(0.016)
Latitude × Reform		0.110***	0.117***
		(0.018)	(0.018)
Ruggedness × Reform		0.0008**	0.0008***
		(0.0003)	(0.0003)
Log Battle Site's Distance to Capital City × Reform		0.281*	0.289*
		(0.148)	(0.151)
Rice × Reform		$-6.84 \times 10^{-5***}$	$-6.79 \times 10^{-5**}$
		(2.46×10^{-5})	(2.62×10^{-5})
Millet × Reform		$-9.02 \times 10^{-5***}$	$-9.88 \times 10^{-5***}$
		(3.27×10^{-5})	(3.41×10^{-5})
Wheat × Reform		0.0002***	0.0002***
		(5.23×10^{-5})	(5.55×10^{-5})
Log Battle Site's Distance to Province × Reform		0.010	0.007
		(0.034)	(0.032)
Province FE	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes
Enemy Type FE	No	No	Yes
Observations	434	434	434
R-squared	0.27	0.35	0.36
Mean of Outcome	0.627	0.627	0.627

Notes: Robust standard errors in parentheses are clustered at province level. ***, **, * denote significance at 1%, 5%, 10% levels.

Table B21. Reduced-Army Robustness: Battles and Lost Battles in the Muzong Window (820–822 CE)

	Rebellion				Mutiny				Uprising			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
NumCCAPref × Reform	-0.0058**	-0.0071***	-0.0058**	-0.0068***	0.0118**	0.0168**	0.0118**	0.0169**	0.0200***	0.0171***	0.0200***	0.0171***
	(0.0025)	(0.0025)	(0.0025)	(0.0024)	(0.0056)	(0.0076)	(0.0055)	(0.0076)	(0.0047)	(0.0041)	(0.0047)	(0.0041)
Agricultural Disasters	0.0005	0.0002	0.0008	0.0006	0.0008	0.0002	0.0011	0.0005	0.0030	0.0039	0.0030	0.0039
	(0.0012)	(0.0014)	(0.0013)	(0.0015)	(0.0025)	(0.0039)	(0.0025)	(0.0039)	(0.0025)	(0.0034)	(0.0025)	(0.0034)
Longitude × Reform	0.0004	0.0003	0.0004	0.0003	0.0002	-0.0016	0.0002	-0.0017	0.0013	-0.0005	0.0013	-0.0005
	(0.0007)	(0.0011)	(0.0007)	(0.0011)	(0.0011)	(0.0020)	(0.0011)	(0.0020)	(0.0009)	(0.0020)	(0.0009)	(0.0020)
Latitude × Reform	-3.91×10^{-5}	0.0007	7.02×10^{-5}	0.0006	-0.0014	-0.0033	-0.0013	-0.0033	0.0006	-0.0020	0.0006	-0.0020
	(0.0006)	(0.0011)	(0.0006)	(0.0011)	(0.0010)	(0.0033)	(0.0010)	(0.0033)	(0.0010)	(0.0019)	(0.0010)	(0.0019)
Ruggedness × Reform	2.43×10^{-5}	-2.11×10^{-5}	2.53×10^{-5}	-2.17×10^{-5}	1.17×10^{-5}	6.34×10^{-5}	1.21×10^{-5}	6.25×10^{-5}	1.17×10^{-5}	-2.99×10^{-6}	1.18×10^{-5}	-2.98×10^{-6}
	(2.31×10^{-5})	(3.08×10^{-5})	(2.32×10^{-5})	(3.05×10^{-5})	(3.86×10^{-5})	(4.27×10^{-5})	(3.85×10^{-5})	(4.23×10^{-5})	(2.81×10^{-5})	(3.85×10^{-5})	(2.81×10^{-5})	(3.85×10^{-5})
Log Distance to Capital × Reform	0.0053	-0.0022	0.0058	-0.0017	0.0021	0.0148	0.0027	0.0154	0.0073	0.0216**	0.0073	0.0216**
	(0.0047)	(0.0077)	(0.0046)	(0.0075)	(0.0083)	(0.0157)	(0.0082)	(0.0158)	(0.0059)	(0.0087)	(0.0059)	(0.0087)
River Density × Reform	-0.0028	0.0160	-0.0043	0.0201	-0.0176	-0.0434	-0.0176	-0.0383	0.1662***	0.1435**	0.1659***	0.1435**
	(0.0497)	(0.0656)	(0.0492)	(0.0649)	(0.1243)	(0.1419)	(0.1264)	(0.1421)	(0.0572)	(0.0616)	(0.0572)	(0.0617)
Log Population × Reform	0.0012	-0.0007	0.0008	-0.0012	0.0011	0.0006	0.0008	0.0002	-0.0031	-0.0019	-0.0031	-0.0019
	(0.0023)	(0.0027)	(0.0023)	(0.0026)	(0.0052)	(0.0077)	(0.0054)	(0.0079)	(0.0033)	(0.0036)	(0.0033)	(0.0036)
Battles × Muzong (820–822)	0.0940	0.1257			0.1001	0.1238*			-0.0079	-0.0002		
	(0.0670)	(0.0806)			(0.0624)	(0.0659)			(0.0082)	(0.0024)		
Lost Battles × Muzong (820–822)			0.4989***	0.5004***			0.4539***	0.4264***			-0.0236	0.0015
			(0.0016)	(0.0020)			(0.0191)	(0.0623)			(0.0158)	(0.0054)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region-Year FE		Yes		Yes		Yes		Yes		Yes		Yes
Observations	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785	4,785
R-Squared	0.06811	0.28799	0.08553	0.30222	0.05573	0.23069	0.06244	0.23525	0.12089	0.40916	0.12089	0.40916
Within R-Squared	0.01297	0.01653	0.03141	0.03618	0.01027	0.01136	0.01730	0.01722	0.01176	0.00754	0.01176	0.00754

Notes: This table tests whether the Reform's effects survive controlling for the troop-reduction channel identified by Peterson (1979). Emperor Muzong issued a secret edict in 820 CE ordering provinces not to refill soldiers lost through attrition; the policy was rescinded in 822 CE. We define $Muzong = 1$ if year $\in \{820, 821, 822\}$ and zero otherwise, and add interactions of *Muzong* with *Number of Battles* and *Number of Lost Battles* to the main DiD specification. Columns 1, 3, 5, 7, 9, and 11 add geographic and disaster controls; columns 2, 4, 6, 8, 10, and 12 further add region-year fixed effects. "Lost Battles" refers to battles that the Tang did not win. Robust standard errors in parentheses are clustered at the province level. ***, **, * denote significance at 1%, 5%, 10% levels.

C Additional Details and Analysis

C.1 Data Sources

The two official histories are the *Old Book of Tang (Jiu Tangshu)* and the *New Book of Tang (Xin Tangshu)*, both of which are part of the "Twenty-Four Dynastic Histories" collection:

- The *Old Book* was compiled in 945 CE under the Later Jin Dynasty (936 - 947 CE), and mainly consulted materials produced by chroniclers employed by the Tang emperors, as well as imperial documents, oral histories of elite families and private histories written by renowned literati.

- The *New Book* was compiled in 1060 CE under the Northern Song consulting similar sources, and differed from the *Old Book* mainly in narrative, institutional details, and the referencing of original materials (Huang 2015: 7-9, 21-26). The two books are used in conjunction by historians of Tang.

The three imperially-commissioned works are the *Comprehensive Mirror in Aid of Governance (Zizhi Tongjian)*, the *Institutional History of the Tang (Tang Huiyao)*, and *Cefu Yuangui*:

- The *Comprehensive Mirror*, published in 1084 CE, is a chronicle of Chinese history from 403 BCE to 959 CE. Its section on Tang consulted similar sources as the two *Books*, and arguably exceeds both of them in quality.

- The *Institutional History* describes the laws and regulations of the Tang administration. It was compiled in 961 CE, and subsumed two works of institutional histories written by Tang scholars. It contains valuable institutional details on mid-to-late Tang.

- *Cefu Yuangui*, compiled in 1013 CE, is an encyclopedia on Chinese political history up to the Tang dynasty. Its section on Tang used similar sources as the two *Books*, and can be considered as the "most comprehensive repository of historical materials from the Tang" (Huang 2015: 54, 68, 248).

The two complementary works are the *Complete Prose of the Tang (Quan Tangwen)* and its addendum *(Quan Tangwen Buyi)*, and excavated tomb epitaphs of Tang political elites:

- The *Complete Prose* is a collection of prose writings from the Tang, compiled in the Qing Dynasty. It contains political writings by renowned officials describing political events they experienced while in office.

- The epitaphs are stone inscriptions of the biographies of the deceased, which are often used as primary sources to study the relevant historical periods (e.g., Tackett 2014; Wen, Wang and Hout 2024). They are assembled by the China Stone Inscription Database.

C.2 Details of Fiscal Policies

We empirically address two fiscal policies from before and during Emperor Xianzong's reign that were not directly related to the CCAs. The first is the fiscal centralization reform in 809 CE. Since the mid-8th century, prefectures had been submitting revenues to both the central government and the commissioner-controlled provincial government. Under the policy of 809 CE, only the headquarters prefecture would pay revenue to the province, and outer prefectures would only submit revenue to the center. This policy reduced provincial revenues, with larger losses expected in provinces with more prefectures under their jurisdiction (Peterson 1979: 526-527; Lewis 2012: 65). To account for its impact, we include an interaction term

between the number of prefectures in each province in 809 CE and the Reform dummy in the regression, and results stay robust (Table B3).

We also consider the "Two-Tax Policy" in 780 CE, which formally allocated tax quotas to provinces (Lewis 2012: 61). Its complete implementation was only limited to several southeastern provinces. We code a dummy variable representing these fiscal basket provinces following Peterson (1979: 520),[50] and interact it with the Reform dummy in our specifications. Results remain robust (Table B4). More generally, Zhang (2010: 44-45) classifies all Tang provinces into four types, one of which captures the aforementioned fiscal basket provinces.[51] We create a categorical variable for these four types and interact it with the Reform dummy in our regressions, and our results stay qualitatively unchanged (Table B2).

C.3 Reform and Commissioner Characteristics

We draw on the extensive biographic and family background information from our newly digitized, original database of 1,450 Tang commissioners. We construct a wide range of variables that capture the proximity of commissioners to the central government in terms of shared interest or identity, and aggregate them across commissioners to our province-year panel data. They include: (1) number of appointments of non-Chinese commissioners;[52] (2) number of commissioners who had civil examination (*keju*) degrees; (3) number of commissioners who entered officialdom through military merit; (4) number of commissioners whose father was a government official; (5) number of commissioners whose father was a

[50]Peterson's map identifies areas that regularly remitted taxes to the central government in 807 CE. We use regular remittance as an indicator of the Two-Tax Policy's effective implementation.

[51]The other three types are the "northeastern warlord provinces", the "central protection provinces" and the "border defense provinces".

[52]The Tang Dynasty employed people who originated from foreign backgrounds as warriors, referred to as "barbarian generals" (*fan jiang*).

government official, and who were not from the local province; (6) number of commissioners whose father was a government official, and who were not centrally-appointed locals.

Descriptively, our data suggests that after Xianzong's reign, the characteristics of appointed commissioners became more "centralized" in various ways. On average, (2), (4), (5), and (6) all increased after 820 CE (the death year of Xianzong), while (1) and (3) declined. These quantitative trends are highly consistent with historians' consensus that, throughout the 9th century and especially after Emperor Xianzong, Tang provincial commissioners became more "civil" both in their career trajectory and cultural identity. They also tended to come increasingly from families with interests tied to the capital (e.g., Zhang 2010; Tackett 2014; Li 2015; Lu 2016; Qiu 2018). Notably, the appointment of *Keju* degree holders as commissioners rose by 123% from 820 CE to the end of the Tang, a trend that aligns with new findings about the rising importance of *Keju* across the Tang Dynasty, particularly in the 9th century (Wen, Wang and Hout 2024).

Table B17 presents results from the DD design, using these seven variables as outcomes. It suggests that most of these personnel changes were not driven by the Reform *per se*. The Reform's effect was statistically insignificant for all outcomes except for the rulers' ability to reappoint commissioners elsewhere. Further analysis using the centrally-mandated reappointment as the "mediator" in the ACDE model shows, again, that the ACDE closely resembles the total effects (left panel of Appendix Figure A3).[53]

[53]The sensitivity analyses for the centrally-mandated reappointments reveal more constrained regions for ρ compared to the ACDE analyses for the uprising results previously discussed. However, this issue is not a major concern because centrally-mandated reappointments, by themselves, do not appear to affect rebellion, mutiny, or uprising (Table B18).

C.4 Qualitative Evidence for the Mechanisms

This section presents a qualitative case study drawn from primary historical evidence to illustrate how communication and coordination problems often impaired the effectiveness of the Tang army following the Reform.

The Pang Xun Uprising

We focus on the initial failure to suppress the renowned Pang Xun Uprising (庞勋起义) of 868–869 CE, a significant uprising often considered the true precursor to the Huang Chao Uprising, which ultimately destroyed the Tang dynasty (Somers 1979: 700). The uprising was triggered when the government refused to let soldiers, who had completed their rotation duty on China's southwestern frontier, return to their hometown of Xuzhou Prefecture. Pang Xun emerged as the leader of these soldiers, who chose to force their way home despite knowing it would brand them rebels.[54]

As Pang's rebels advanced toward Xusi Province, its Provincial Commissioner dispatched his headquarters army and ordered forces from Suzhou and Sizhou to intercept the rebels. Successful interception required close coordination between these forces. Commander Yuan's headquarters army attempted an ambush, but the rebels detected the plan and evaded it. Yuan's forces then withdrew and delayed pursuit until the following morning. Without timely support from the headquarters army, Suzhou's isolated prefectural force was defeated, allowing the rebels to capture the undefended prefectural capital. The failure to coordinate the separately deployed forces thus turned an unsuccessful ambush into a major defeat.[55]

[54] *Old Book of Tang* (Vol. 177) records the postponement of the soldiers' replacement and their subsequent selection of Pang Xun as commander: "旧三年一代，至是戍卒求代。尹戡以军帑匮乏，难以发兵，且留旧戍一年...戍卒怒...立粮料判官庞勋为都将。"

[55] *Comprehensive Mirrors* (Vol. 251). The original Chinese text reads: "仍命宿州出兵苻离，泗州出兵于

This disastrous defeat dramatically shifted the conflict's momentum, enabling the rebels to recruit numerous commoners drawn by their success. Capitalizing on this surge, the rebels subsequently defeated Commander Yuan's headquarters army, captured the provincial headquarters of Xusi, and captured (and later executed) the provincial commissioner of Xusi.

The catastrophic loss was evidently caused by inadequate coordination between Commander Yuan's forces (directly under the Commissioner of Xusi) and the prefectural armies. Once the ambush plan was compromised, Yuan's forces hesitated to engage the rebels in open combat.[56] Had the armies from Prefectures Suzhou and Sizhou fully cooperated from the beginning, the Tang forces likely would have been able to effectively confront the rebels.

The failed ambush in Xusi also sheds light on an earlier decision by the Commissioner of Huainan, whose territory the rebels crossed on their way home. Although one of his generals advised a sophisticated ambush, the Commissioner chose to avoid conflict, allowing the rebels to advance unopposed toward their hometown.[57] While historically criticized as cowardice (Somers 1979: 699), this decision can plausibly be understood, within our theoretical framework, as reluctance to undertake an ambush that demanded precise coordination and communication—capabilities that were substantially undermined by the Reform.

These coordination failures cast a different light on the seemingly "heroic" defense led by Du Tao, Prefect of Sizhou Prefecture, when his city was besieged by rebels after they had

虹以邀之...复引兵退宿于城南，明旦，乃进追之。时贼已至苻离，宿州戍卒五百人出战于濉水上，望风奔溃...再宿，官军始至，贼守备已严，不可复攻。"

[56] *Comprehensive Mirrors* (Vol. 251). The original Chinese text reads: 日暮，贼至任山，馆中空无人，又无供给，疑之，见负薪者，执而榜之。果得其情，乃为偶人执旗帜列于山下而潜遁。比夜，官军始觉之，恐贼潜伏山谷及间道来袭，复引兵退宿于城南，明旦，乃进追之。

[57] *Old Book of Tang* (Vol. 172). The original Chinese text reads: 都押衙李湘白曰："徐兵擅还，必无好意。虽无诏命除讨，权变制在方...计其水路，须出高邮县界，河岸斗峻而水深狭。若出奇兵邀之，俾获船纵火于前，劲兵奋击于后，败走必矣。若不于此诛锄，俟济淮、泗，合徐人负怨之徒，不下十万，则祸乱非细也。"性懦缓，又以不奉诏命，谓湘曰："长淮已南，他不为暴。从他过去，余非吾事也。"

already captured the provincial commissioner and overrun other cities. The records describe Du's army as "fully prepared" and detail his valiant resistance against overwhelming odds as the Tang soldiers were clearly outnumbered.[58] Yet the irony is clear: Sizhou Prefecture's army had been ordered from the outset to join Commander Yuan's forces but evidently failed to do so. Du Tao's forces remained capable of defending their own city, yet failed to provide effective support to Suzhou or the provincial capital, even as the provincial commissioner appealed for reinforcements.[59] This contrast betrays the failure to coordinate the province's military resources. The historical account devotes considerable space to praising Du Tao's defense, even though it implicitly reveals an earlier lack of cooperation with other Tang forces. This episode shows a striking breakdown in the provincial command structure following the Reform.

Ultimately, the Tang dynasty had to enlist the aid of northern nomadic forces to suppress Pang Xun's uprising. After the rebellion, the imperial court endorsed a statement blaming failures of command for the recurrent disorder: "Although Xuzhou has repeatedly experienced disorder, its inhabitants are not necessarily all wicked and rebellious; rather, failures of command have allowed malicious elements to exploit opportunities."[60]

Peterson (1979: 554, 559) places these failures in the context of reforms that divided

[58] *Comprehensive Mirrors* (Vol. 251) records Du's defensive preparations and the rebels' unsuccessful assault despite their numbers exceeding ten thousand: 泗州刺史杜伷勋作乱，完守备以待之……勋以泗州当江、淮之冲，益发兵助圆攻之，众至万馀，终不能克。 The *New Book of Tang* (Vol. 166) likewise emphasizes his repaired fortifications and complete military equipment: 之闻难，完城隍，阅器械无不具。

[59] *Comprehensive Mirrors* (Vol. 251) records the initial orders for both prefectures to intercept the rebels: 仍命宿州出兵苻离，泗州出兵于虹以邀之。 After Suzhou fell and Yuan's army was defeated, the commissioner appealed to neighboring provinces for help: 乙亥，庞勋引兵北渡濉水，逾山趣彭城。其夕，崔彦曾始知元密败，移牒邻道求救。 The provincial capital nevertheless fell, while Sizhou subsequently mounted a determined defense of its own. This sequence exposes the failure to combine local military resources in defense of the province.

[60] *Comprehensive Mirrors* (Vol. 252). The original Chinese text reads: 徐州虽屡构祸乱，未必比屋顽凶；盖由统御失人，是致奸回乘衅。

military authority between governors and prefects and weakened provincial forces' ability to maintain local security. In his discussion of the Pang Xun Uprising, Zhang (2011: 109) likewise observes that prefectural armies could constrain commissioners but struggled to fight independently. The same division of command that protected the emperor from powerful provincial commissioners also weakened their ability to suppress rebellion.

D Additional Historical Context

D.1 Connections with the historical literature

The Tang Dynasty after the An Lushan Rebellion of 755 CE has been extensively researched by historians. Contrary to traditional narratives that depict the post-755 Tang as chaotic and weak, recent "revisionist" accounts increasingly highlight stability in this era. These accounts emphasize enhanced authority over previously rebellious provinces, especially following Emperor Xianzong's reign (805-820 CE) (e.g., Li 2015; Lu 2016; Qiu 2018). They also recognize that, although the central government substantially strengthened its control over the provinces, the dynasty as a whole witnessed a gradual erosion of its grip over society, particularly among lower-level officers and soldiers prone to mutiny (Qiu 2018: ch. 6). The first view is consistent with our finding that the military centralization reform reduced the incidence of commissioner rebellions, and the second is consistent with our finding that the Reform led to increased soldier-led mutinies against the commissioners.

D.2 CCAs and the An Lushan Rebellion

This section explains that variation in the CCAs stemmed primarily from military activities during the An Lushan Rebellion (755–763 CE), which were themselves shaped by the internal politics within the Tang imperial palace.

The CCAs as a response to the An Lushan Rebellion

The An Lushan Rebellion was the severest threat to the dynasty's survival to date, as An's forces sacked the Tang capital, Chang'an, after occupying large swaths of the Chinese hinterland. In response, the Tang rulers allowed the CCAs to be established as a way for the local military powerholders to acquire more power and discretion. Such power and discretion, especially in terms of recruiting and deploying their own troops, were urgently needed to suppress the rebellious activities (Graff 2003: 229; Zhang 2010: 78; Zhang 2011: 101; Li 2015: 380-530).

Though the CCAs were increasingly becoming a problem for political order and therefore to be addressed by the Reform of 819, the necessity of the CCAs during the mid-8th century as a tool to defeat the An Lushan rebels was widely understood by the Tang contemporaries. The exact edict promulgating the Reform of 819 CE that removed the CCAs was even accompanied by a comment explicitly emphasizing how the need for discretion to be granted to provincial commissioners was the root of the CCAs: "Since the onset of the Great Difficulties, military strength has been needed throughout the realm, and the authority of the military commissioners has become especially significant."[61] As shown in Table B16, one

[61] *Cefu Yuangui*, Vol. 60. "Great Difficulties" was a phrase commonly used to refer to the An Lushan Rebellion by the contemporaries in the post-755 Tang. The original Chinese text is: 自艰难以来，天下右武，节将权尤重.

standard deviation in the number of battles during those eight years explains more than 50% of a standard deviation in the CCAs, while most economic and geographic predictors are insignificant in the combined regression.

Idiosyncratic nature of the military activities during the Rebellion years

While the geographic distribution of military activities usually reflects economic and geostrategic differentials across localities, the An Lushan Rebellion and other rebellions that erupted concurrently were heavily influenced by the idiosyncratic nature of the internal politics within the Tang palace. Specifically, the battles that occurred throughout China during the Rebellion years reflected significant contingency and uncertainty in the interactions between the imperial government and An Lushan, as well as among various factions vying for power within the Tang central government, leading up to and during the Rebellion.

A notable feature of the background to An Lushan's rebellion was the deadly succession politics of the Tang dynasty, where emperors often ended up deposing and even killing their own designated successors. Seven out of ten crown princes before 755 CE failed to succeed their father emperors. Among these seven, one was killed by his own brother, one was deposed and then died naturally, and the remaining five were all killed by their own parents, whether they had been deposed before their deaths or not. Among the three crown princes who ultimately became emperors, two did so via palace coups that forced their fathers into abdication. Emperor Xuanzong, reigning at the time of An Lushan's rebellion, was particularly infamous for his suspicion of his kin, having already killed three of his own sons prior to 755 CE, one of whom was the crown prince. Emperor Xuanzong himself led two palace coups against succession rivals when he was just a prince, eventually forcing his father

into abdication.

The new crown prince, designated in 738 CE, had been under strong suspicion from the beginning. Emperor Xuanzong empowered Chief Minister Li Linfu to counterbalance the crown prince (Ren 2024; Ding 2014). Importantly, the struggle between Li and the crown prince directly implicated powerful military commissioners at the imperial frontiers (Wang 2020; Ren 2024). The crown prince had long been a patron to the commissioners at the northern and northwestern commands (Wang 2020), while Li sought to curtail his influence by rapidly promoting An Lushan in the northeastern command to the position of commissioner and later expanding An's sphere of influence to include some provinces in the central north.

This arrangement fell apart when Li died, and Yang Guozhong, a second cousin of Emperor Xuanzong's favorite concubine, became the chief minister. While Yang continued to balance against the crown prince on the Emperor's behalf, he was not An's initial patron like Li, and their relationship soon soured. Within three years of Li's death, An launched a rebellion after Yang executed one of his allies at court, confirming his long-held apprehension that the Emperor and central government ministers never trusted his loyalty (Ren 2024).[62] In short, both the rise of An Lushan to a commissioner with the entire northeastern region under his command and his eventual rebellion were the results of internal politics.

The Tang responses to the Rebellion, which also influenced the geographic distribution of battles during the Rebellion years, continued to reflect maneuvers in deadly succession politics. First, the battle locations extended westward as the rebels continued their march

[62]This mutual suspicion was exacerbated by An's distinct identity, which set him apart from the established political elite in the Tang. He was of Sogdian and Göktürk origin and had spent most of his adult life in the imperial periphery as a warrior, an identity and career trajectory that central government officials held in contempt. Wang (2020) argues that Emperor Xuanzong and Li increasingly turned to non-Chinese warriors in the Tang military to dilute the crown prince's allies in the army, who were mostly Chinese. Meng (2015) argues that An Lushan earned Emperor Xuanzong's trust by openly showing disrespect to the crown prince.

and managed to capture Tong Pass, the last stronghold defending the Tang capital. The loss of Tong Pass was a direct result of power struggles between Yang Guozhong and Emperor Xuanzong on one hand and the crown prince on the other. As the Emperor recalled generals and troops from northern and northwestern commands to defend Tong Pass against the northeastern rebels, he worried about their loyalty due to their connections to the crown prince. In particular, Tang commander Ge Shuhan was widely considered to be the crown prince's key ally and had been on bad terms with Yang, the chief minister tasked with counterbalancing the crown prince (Wang 2020). Both the Emperor and Yang feared that Ge was using the rebels as an excuse to gather all the remaining Tang troops under his command, eventually threatening their positions and aiding the crown prince. Consequently, they compelled Ge and his army to *exit* Tong Pass, arguably the most formidable fortress at the time, and confront the rebels in open battle. This tactical blunder led to the decimation of the best troops and the loss of Tong Pass, along with the capital (Wang 2020; Ren 2024).[63]

Second, due to succession politics, particularly the struggles between the crown prince and his brother Prince Yong, the battles during the Rebellion years extended to the southeast (the Yangtze Delta area). As Emperor Xuanzong fled the capital, he issued an edict granting another son, Prince Yong, command over regions in the upper Yangtze River. The crown prince, seeing his father flee to the southwest, seized the opportunity to lead his entourage to the northwestern region, where his powerbase was located, and declare himself the new emperor, making his father the "retired emperor." Meanwhile, Prince Yong had already arrived in the upper Yangtze River area. However, Prince Yong overstepped by advancing

[63]Both Wang (2020) and Ren (2024) also discuss a more extreme view that Yang wanted the commander to lose and therefore deliberately compelled him to lead the army out of the Pass.

further eastward to establish a powerbase in the Yangtze Delta region with the ambition to eventually take the throne himself. This eastward advancement was neither sanctioned by Emperor Xuanzong's original edict nor approved by the former crown prince, now Emperor Suzong. Nonetheless, the retired emperor Xuanzong tacitly supported Prince Yong in an attempt to balance his sons against each other (Li 2015). The new emperor (Emperor Suzong) acted swiftly to convince local officials and military powerholders to join his side and provoked Prince Yong into an open rebellion, which was brutally crushed.

Importantly, as Emperor Suzong sought to consolidate his control over the Yangtze Delta after Prince Yong's rebellion, his efforts exacerbated tensions between Liu Zhan, a locally stationed general, and several other officials. Liu was eventually compelled to rebel, and Emperor Suzong had to call upon northern troops to defeat him (Li 2015).

Neither the Rebellion of Prince Yong nor the Rebellion of Liu Zhan was part of An Lushan's Rebellion against the Tang, but the battles fought in both instances occurred during the An Lushan Rebellion years, and were consequences of succession politics in the Tang. These military activities were viewed the same way - threats to the empire, and therefore further contributed to the reasoning that provincial military commissioners siding with the imperial government needed to be empowered (through the CCAs).

Third, the battle intensity during the Rebellion years was high in central China also due to internal political concerns. The new Emperor Suzong explicitly ignored advice from a prominent geostrategic adviser that he should focus the war efforts in the north and northeast, targeting An Lushan's original powerbase to eventually exhaust the rebels. Instead, he concentrated military efforts on recovering the Tang capital, Chang'an, and its secondary capital, Luoyang, both located in central China. Historians have long argued that this tactical

decision was driven by political rather than military concerns, as Emperor Suzong urgently needed political victories to overshadow his father, the retired emperor (Wang 2020).

E The Byzantine Parallel

The Byzantine Empire provides a parallel case of military centralization and its costs. Its *themes* system broadly resembled the Tang's provincial military commissioner system. Fear of rebellion made emperors reluctant to entrust generals with forces and resources that could be turned against the throne (Kaegi 1981:299). These concerns were acute under Basil II (r. 976–1025 CE), who faced major rebellions led by Bardas Skleros and Bardas Phokas, supported by eastern provincial armies and their commanders (Holmes 2005). Basil subsequently sought to centralize military authority through tighter control over appointments: he posted generals away from their landed bases and paired commanders so that they could check one another (Treadgold 1997:545). He also extended the existing system of regional commands, or ducates and katepanates, through which senior imperial appointees exercised authority over small frontier themes (Holmes 2005).

This centralized military system imposed costs on coercive capacity. Local commands could meet limited threats, but major invasions required forces to be assembled from the center: "it was still up to the emperor, or one of the two commanders-in-chief, rather than a local commander to assemble an appropriate force" (Haldon 1999:85–90). Dependence on central coordination made simultaneous attacks on different fronts particularly difficult to defend against, as the emperor and the main army could rarely move between them in time (Haldon 1999:90).

The priority given to central control also compromised the defense of Thessalonica, the empire's most important city after Constantinople, during the Norman invasion of 1185. With the city under siege, the emperor Andronikos I assembled forces from the provinces to relieve it. Distrusting these provincial armies, however, he placed these forces under different commanders of his own choosing, "who failed to cooperate" (Treadgold 1997:654). These imperial appointees included his son and a trusted eunuch. Most relief commanders remained far from the besieged city (Choniates 1984:175). The failure of the relief effort helped the Normans capture and sack Thessalonica (Treadgold 1997:654).

As in Tang China, efforts to centralize control over military powerholders undermined the coordination needed to deploy the state's forces effectively. Securing central authority and maintaining coercive capacity were distinct, and sometimes competing, objectives.

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